

Single-Cell RNA Sequencing of Brain and Spleen in a Mouse Tumor Model Across Immunization Conditions

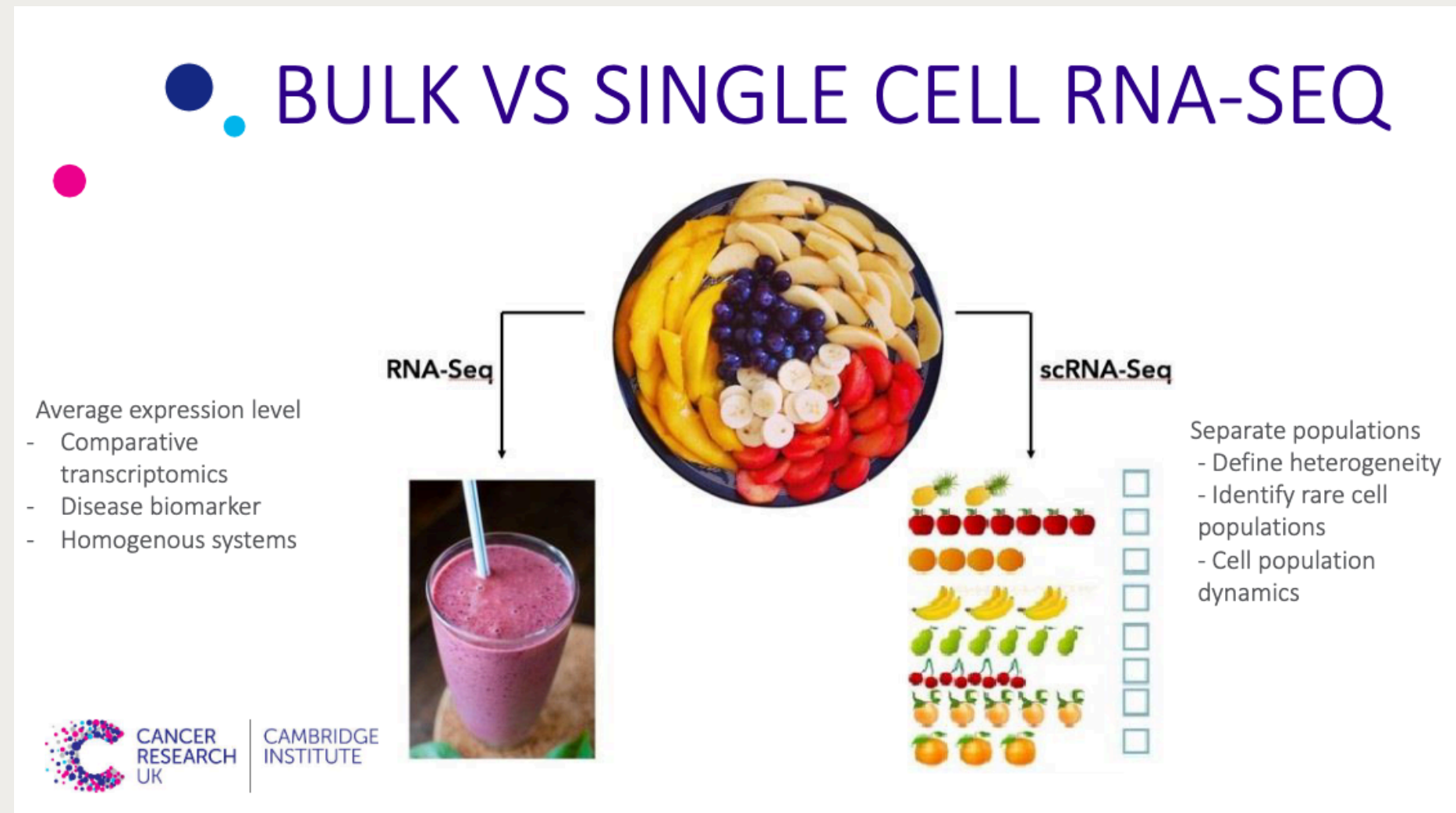
07.06.2026

Presented by:
Kaye Chen



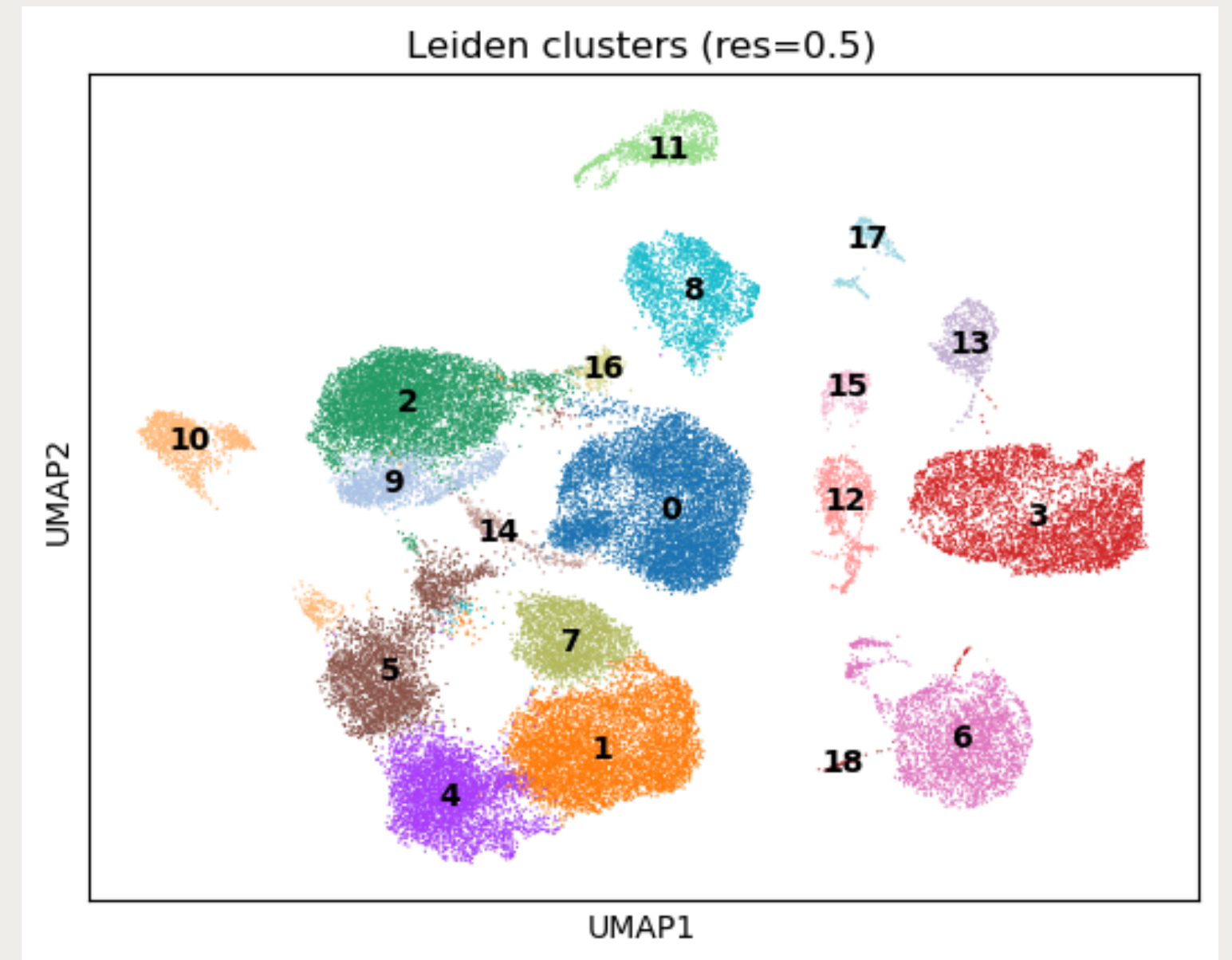
What is scRNA seq?

- RNA is the active readout: it shows which genes a cell is actually using right now
- Bulk RNA-seq averages across all cells → one profile per tissue; single-cell measures each cell individually → thousands of profiles per sample
- scRNA-seq lets us isolate cell profiles, especially the T cell response to the tumor



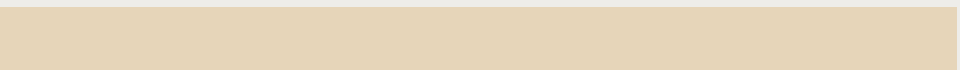
What do we get out of it?

- Each cell gets a gene expression profile
- Cells with similar profiles cluster together
→ we can identify cell types and states
- End result: a map of every major cell population in the tissue



Experimental Overview

Alignment → QC Process → Clustering → Annotation

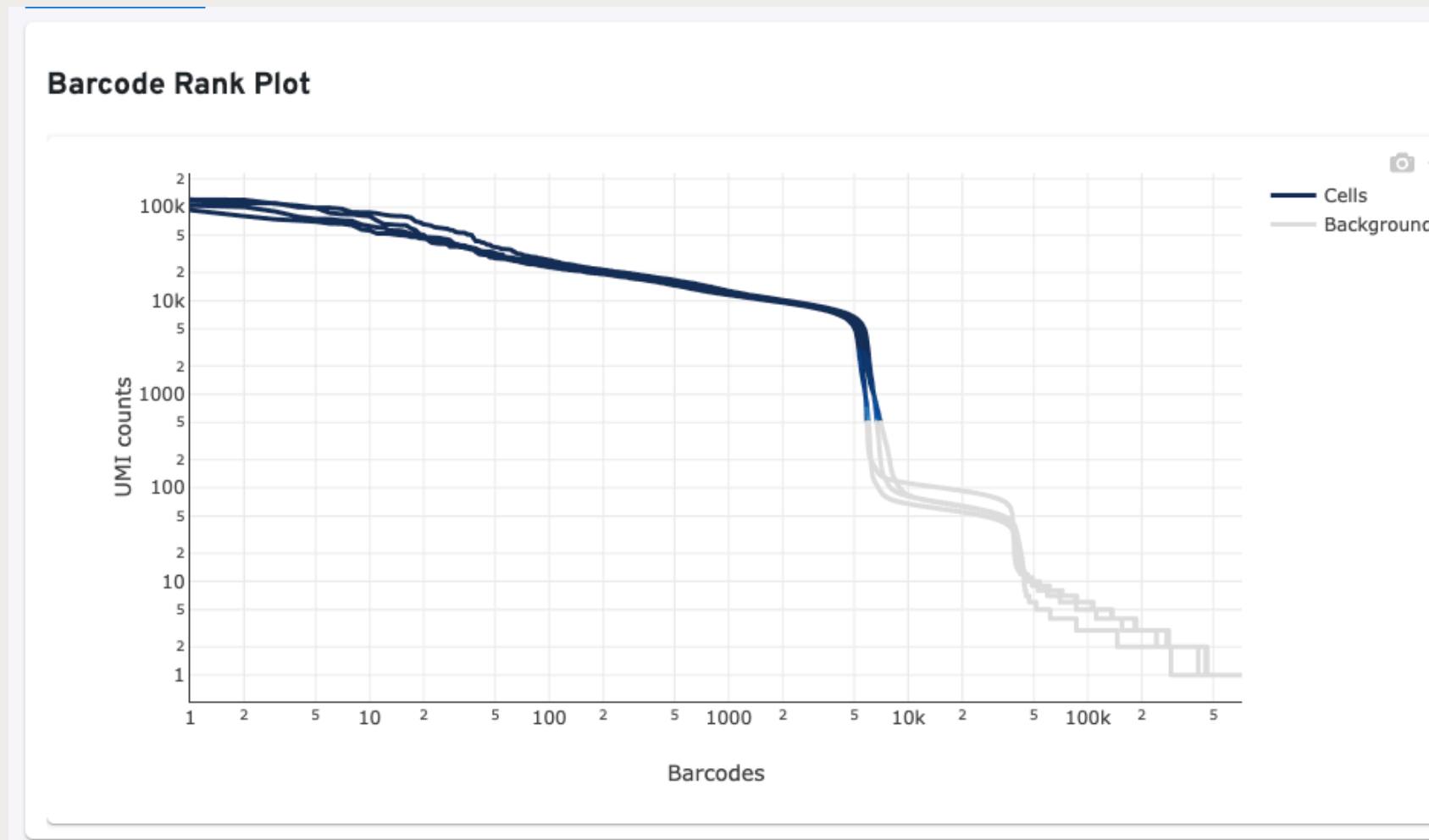
- 4 mouse groups based on:
 - Pre-immunization (yes/no)
 - Tumor type (CT2A-gD vs CT2A-Ctrl)
 - Brain tumor tissue + splenocytes collected from each group
- 

- All samples passed raw read quality checks
- Assessed using FastQC and summarized with MultiQC
- Key metrics looked good

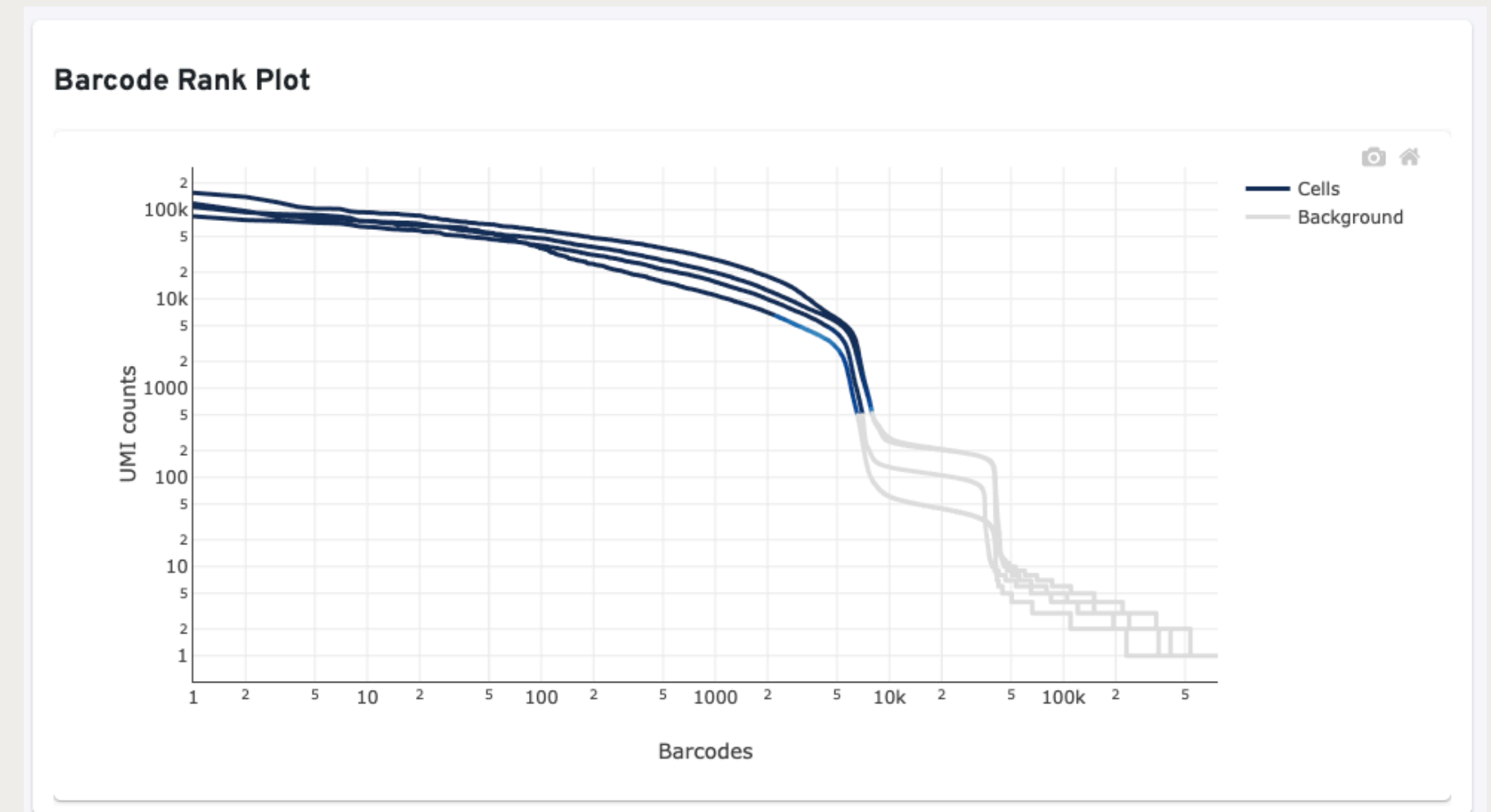
CellRanger Alignment

Reference Genome: Mouse
reference (GRCm39) - 2024-A

Splenocyte Samples



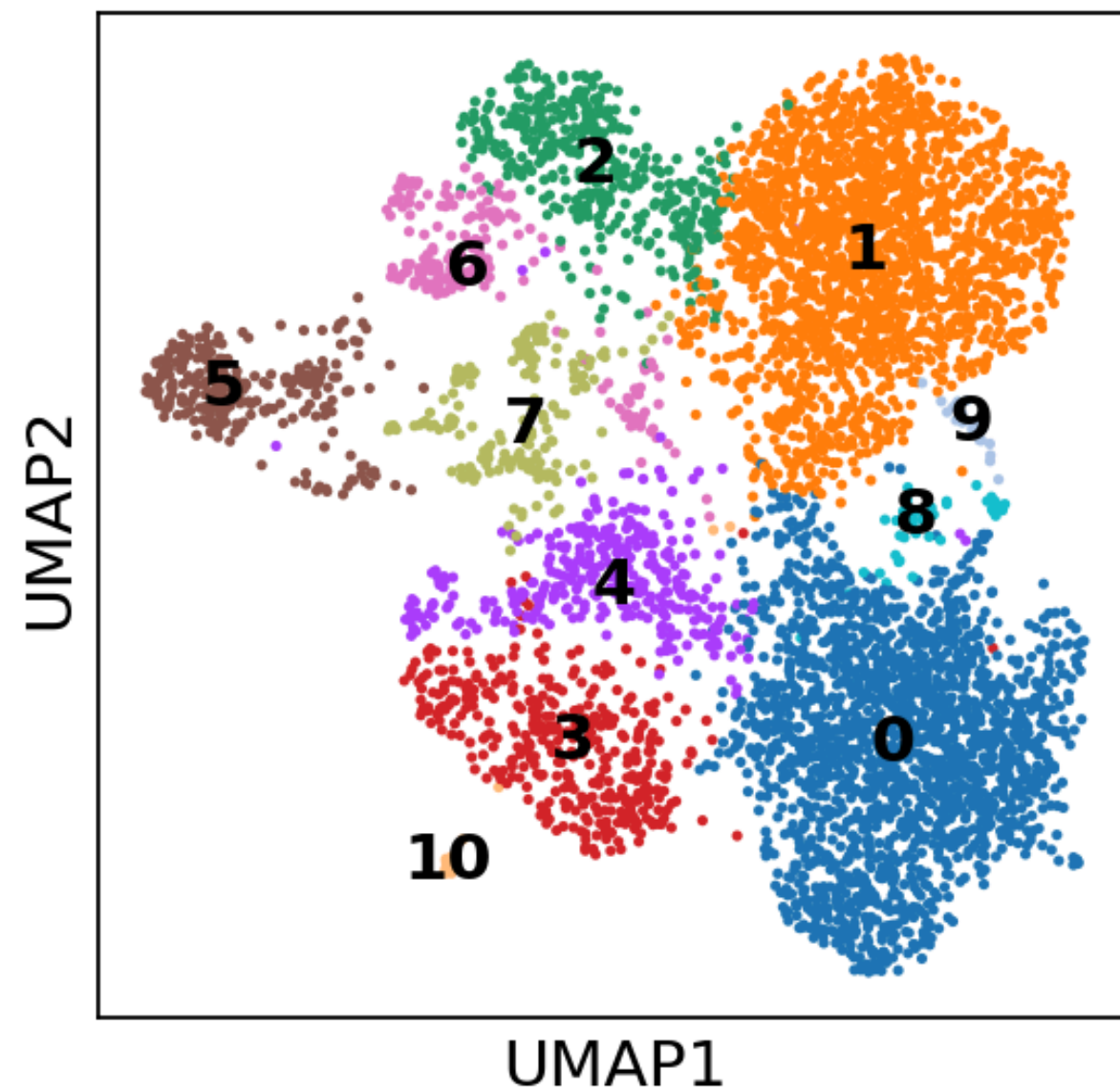
Brain Samples



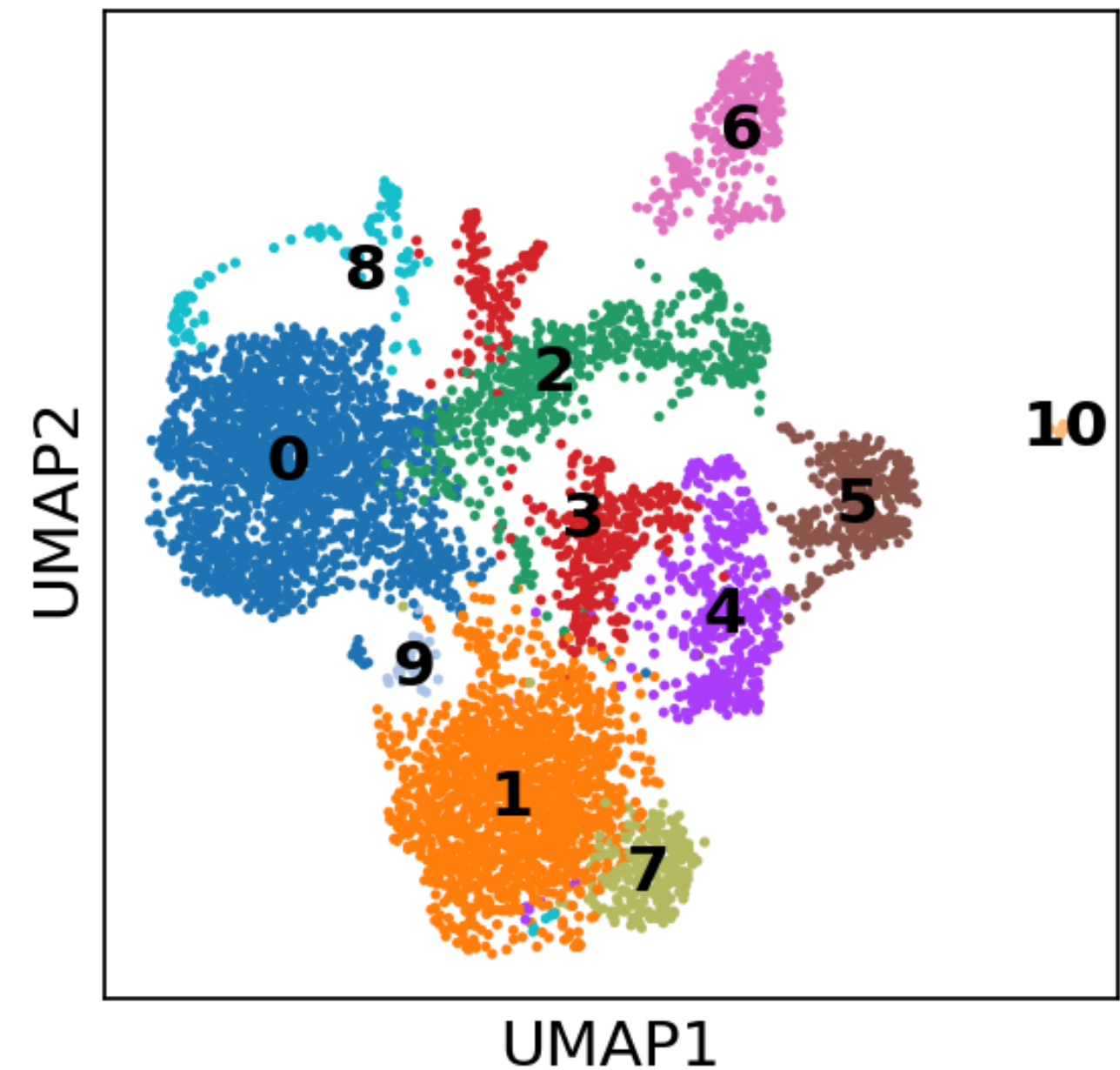
DecontX Ambient RNA Removal

Ambient RNA removal reshapes cluster structure.
One sample example: 11 → 11 clusters with improved separation/tighter clustering

No Ambient RNA Removal - Leiden Clusters

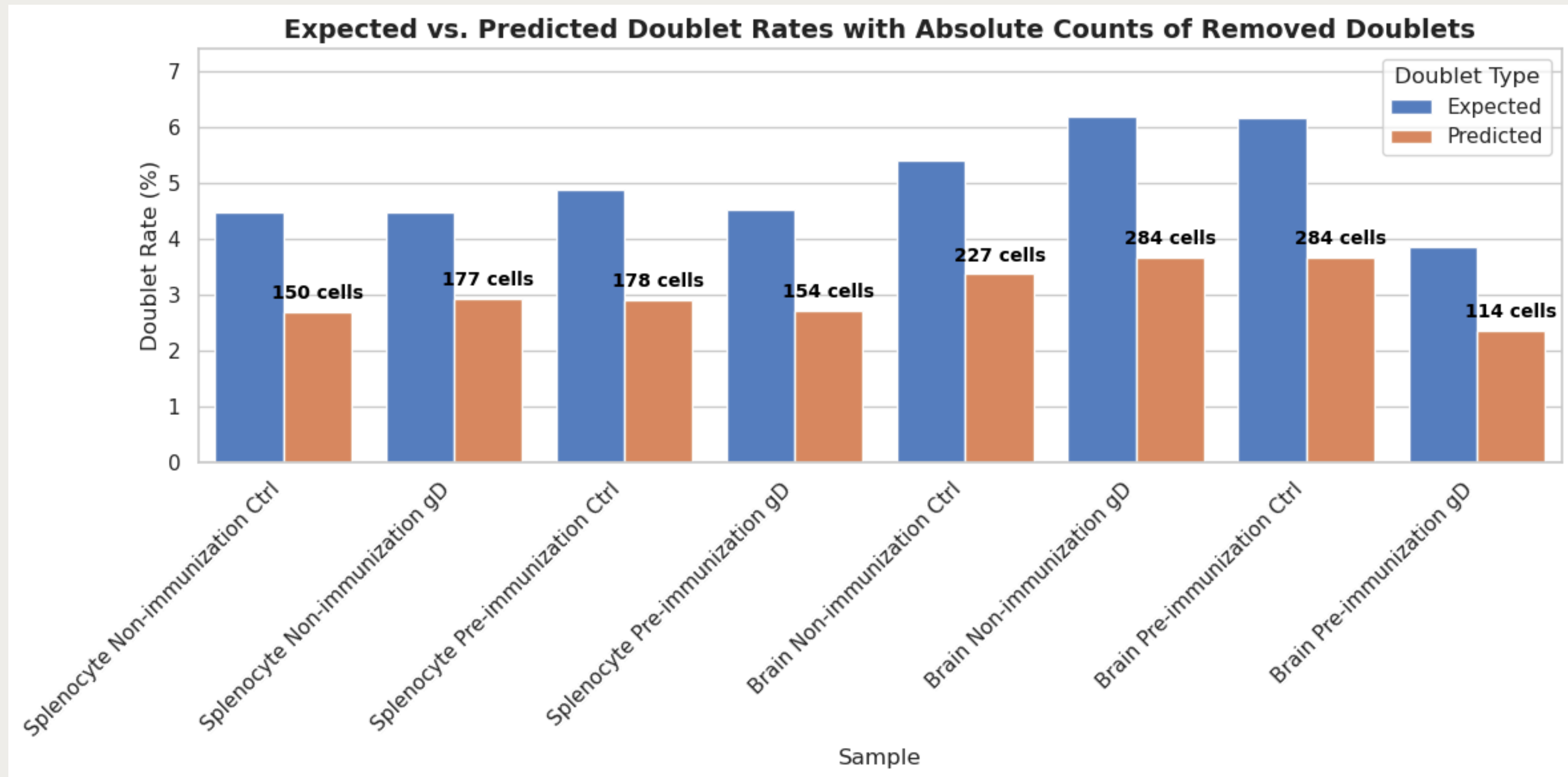


decontX - Leiden Clusters

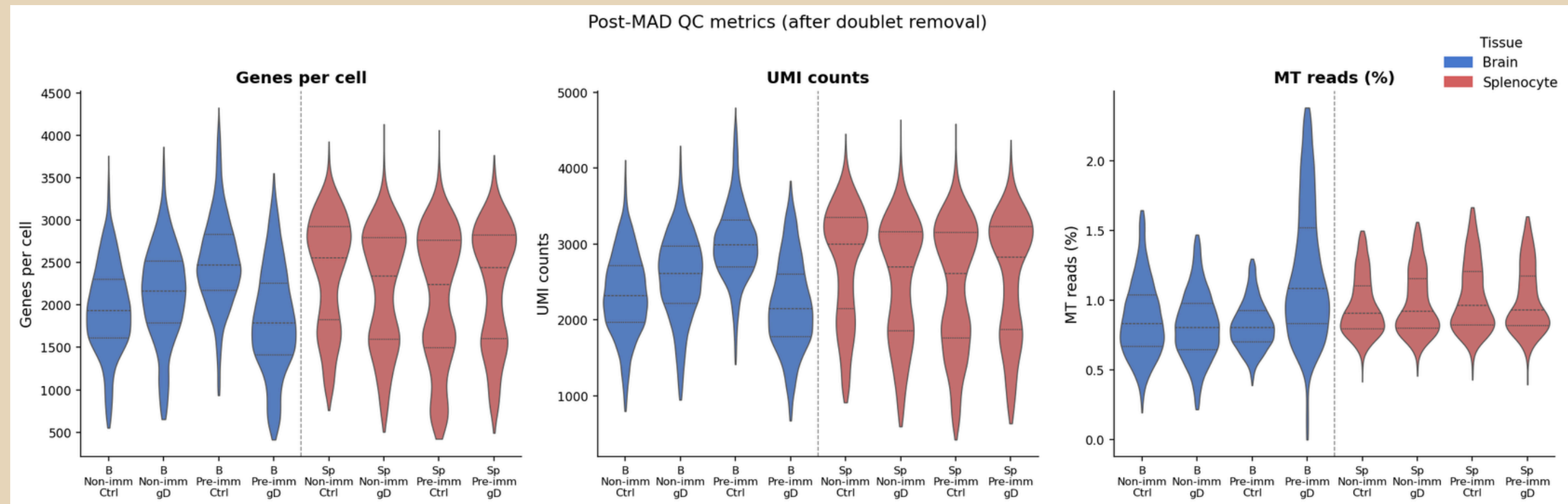


- scDblFinder run per sample prior to clustering
- Predicted rates consistently below expected across all 8 samples

ScDblFinder Doublet Detection



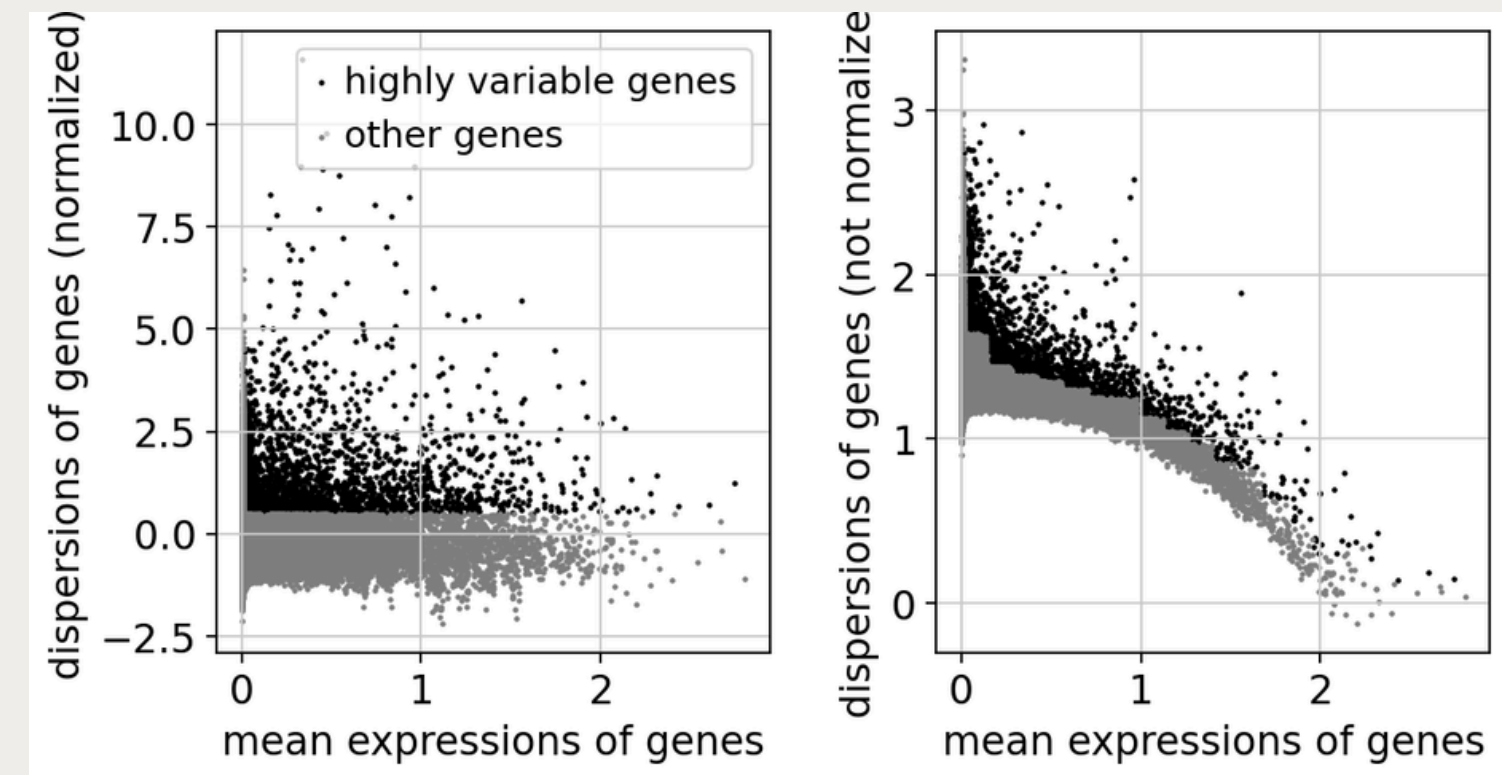
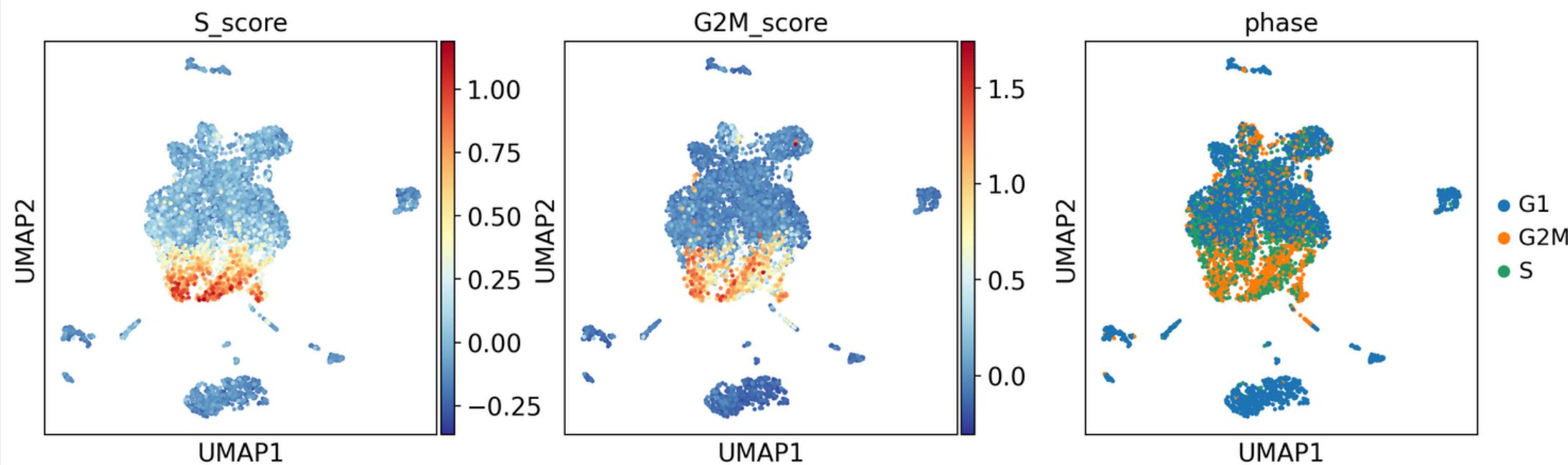
Median Absolute Deviation based thresholds applied consistently across all 8 samples; counts normalized to 1e4 and log1p transformed



- `n_genes_by_counts`: median $\pm$ 5 MADs
- `total_counts`: median $\pm$ 5 MADs
- `pct_counts_mt`: upper bound = min(median + 3 MADs, 10%)

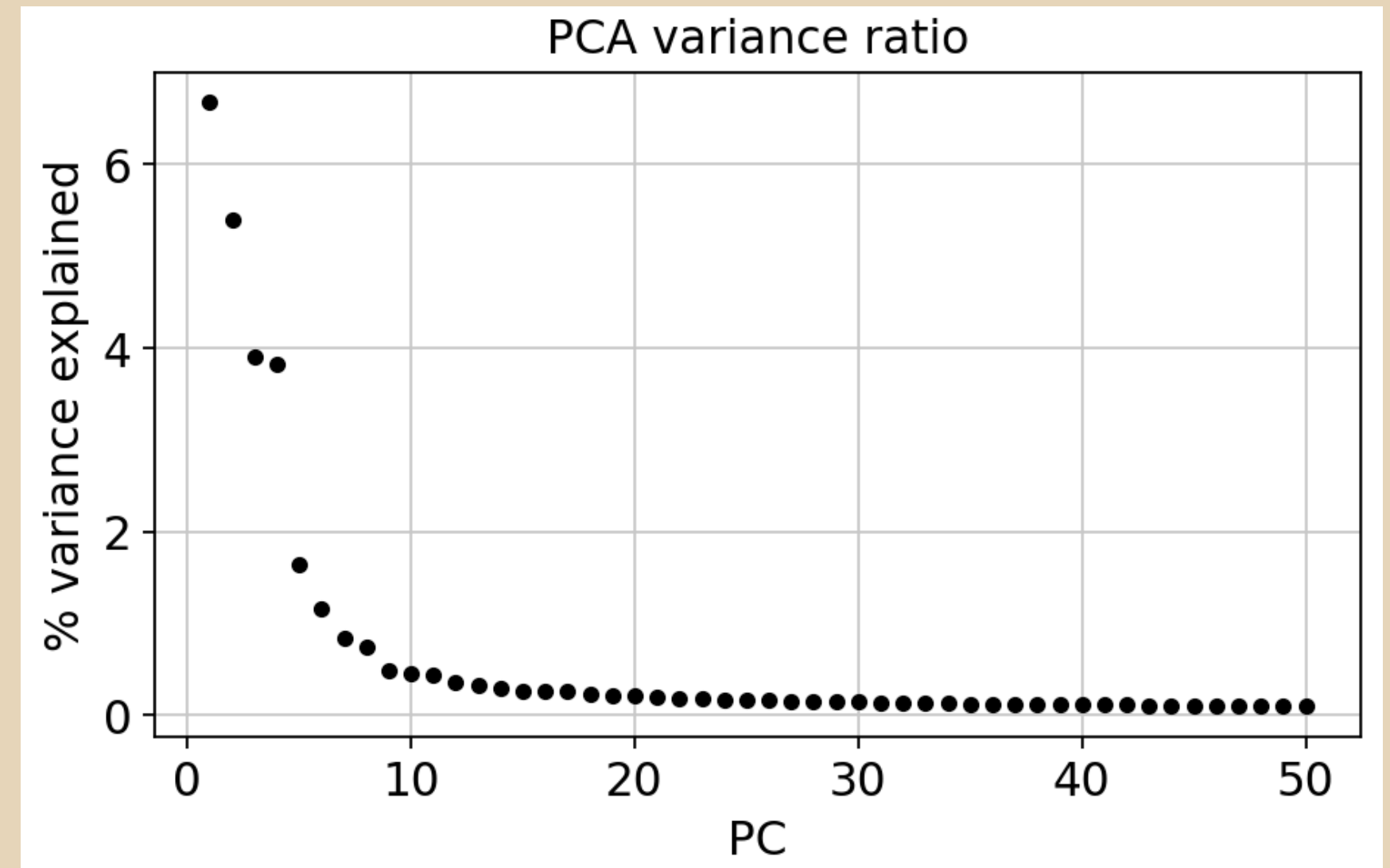
Cell cycle and MT/ribo genes removed prior to HVG selection to isolate biological signal

- S and G2M phase scores calculated using gene reference list
- MT and ribosomal genes excluded from HVG selection
- 2,292–3,686 HVGs selected per sample



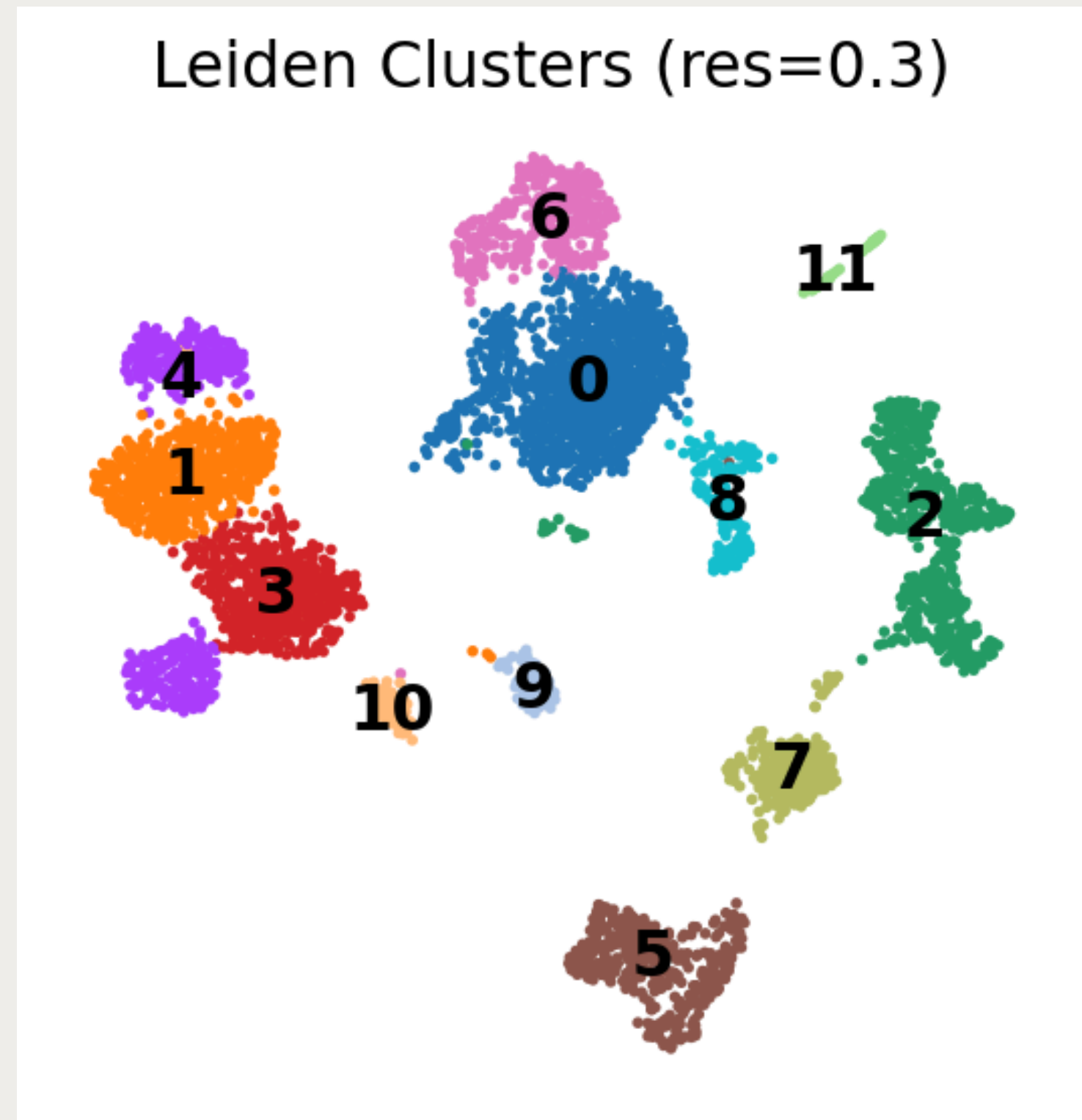
Cell cycle and MT scores regressed out if needed; top 20 PCs retained for downstream analysis

- S and G2M phase scores calculated using gene reference list
- MT and ribosomal genes excluded from HVG selection
- 2,292–3,686 HVGs selected per sample

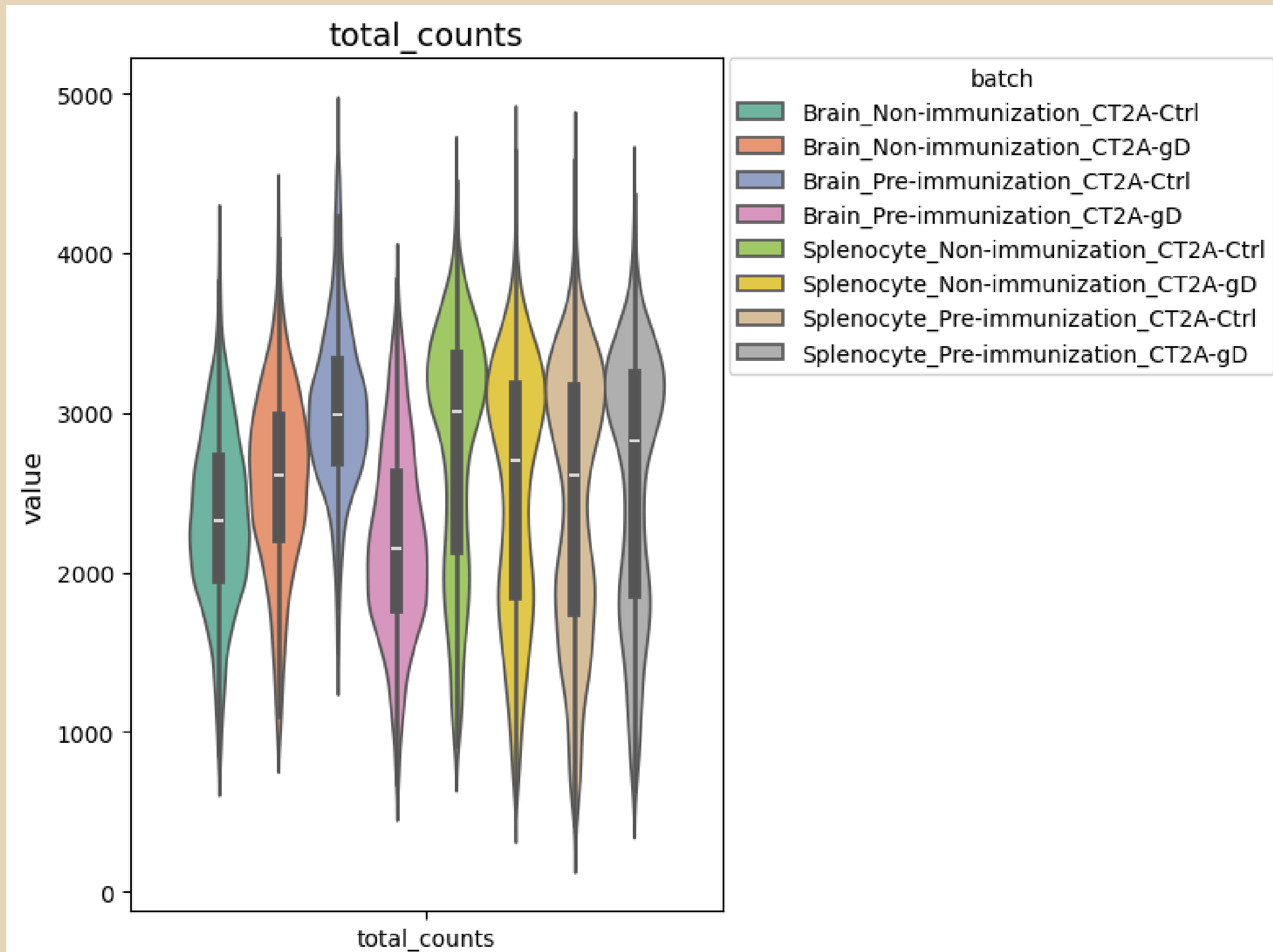


Per-sample clustering confirmed all major populations represent real cell types

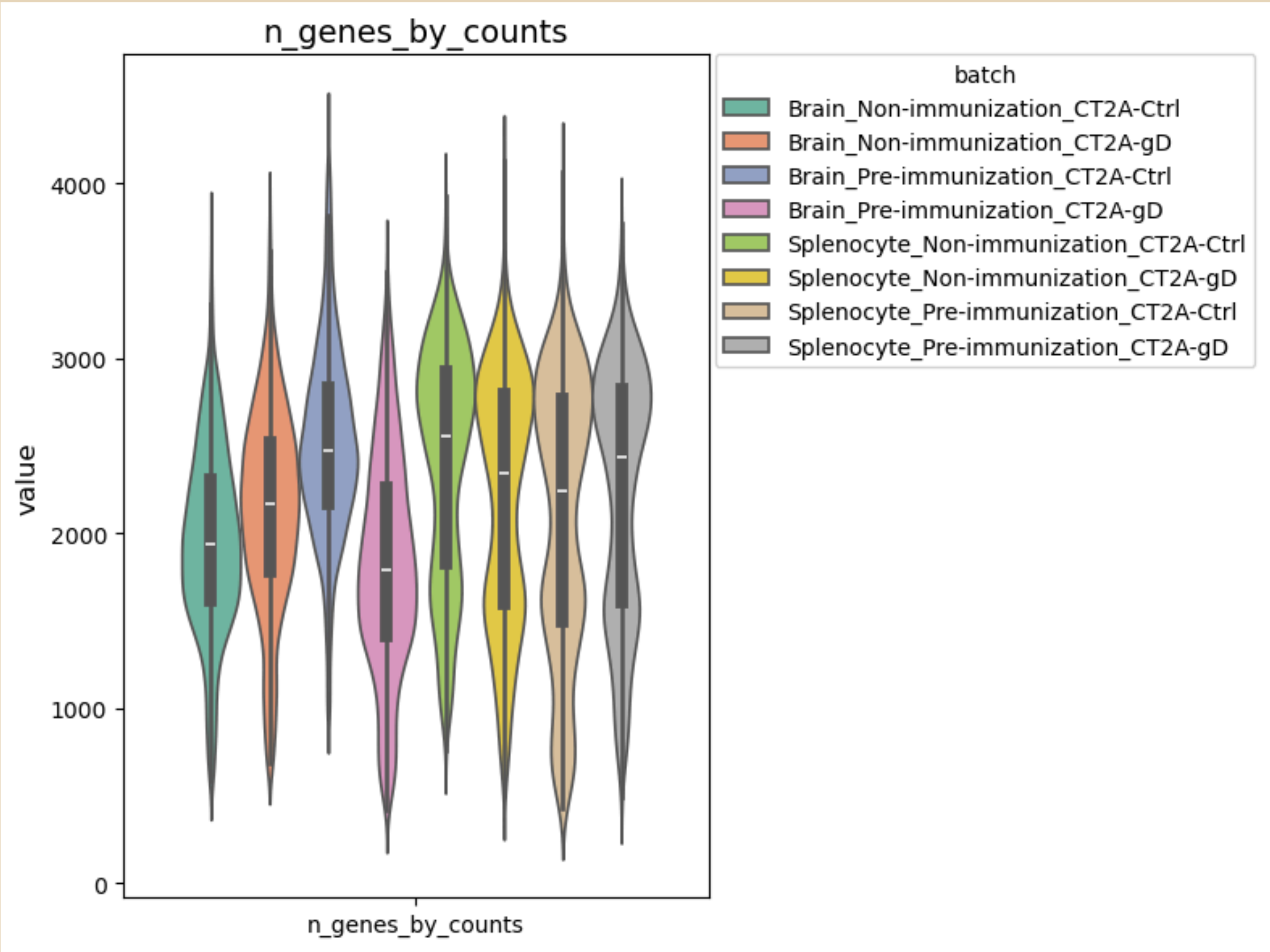
- Leiden clustering run per sample at resolutions 0.2–0.5
- Cluster marker genes examined to confirm real cell type identities



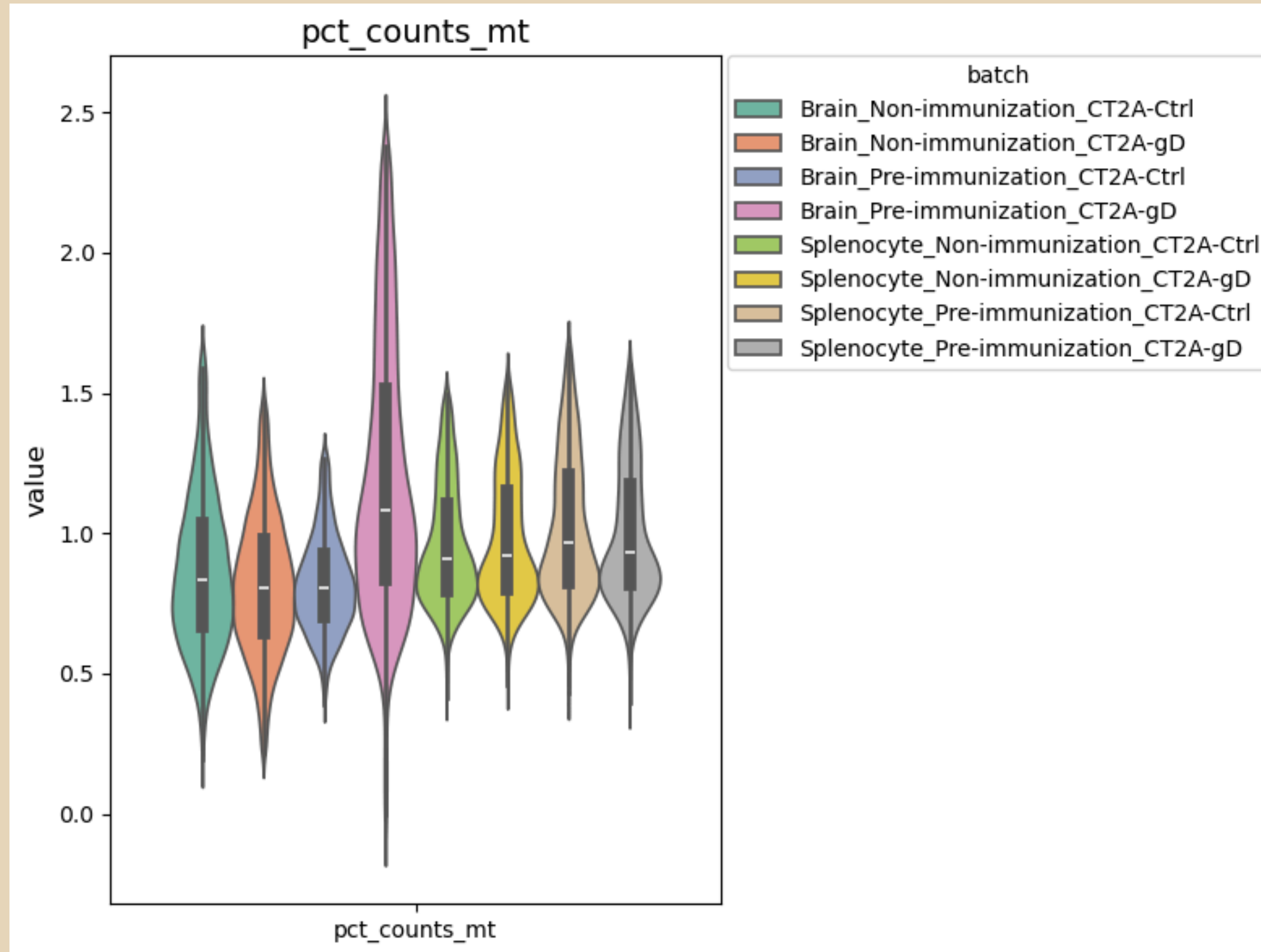
Total UMI counts are consistent across all 8 batches post-QC



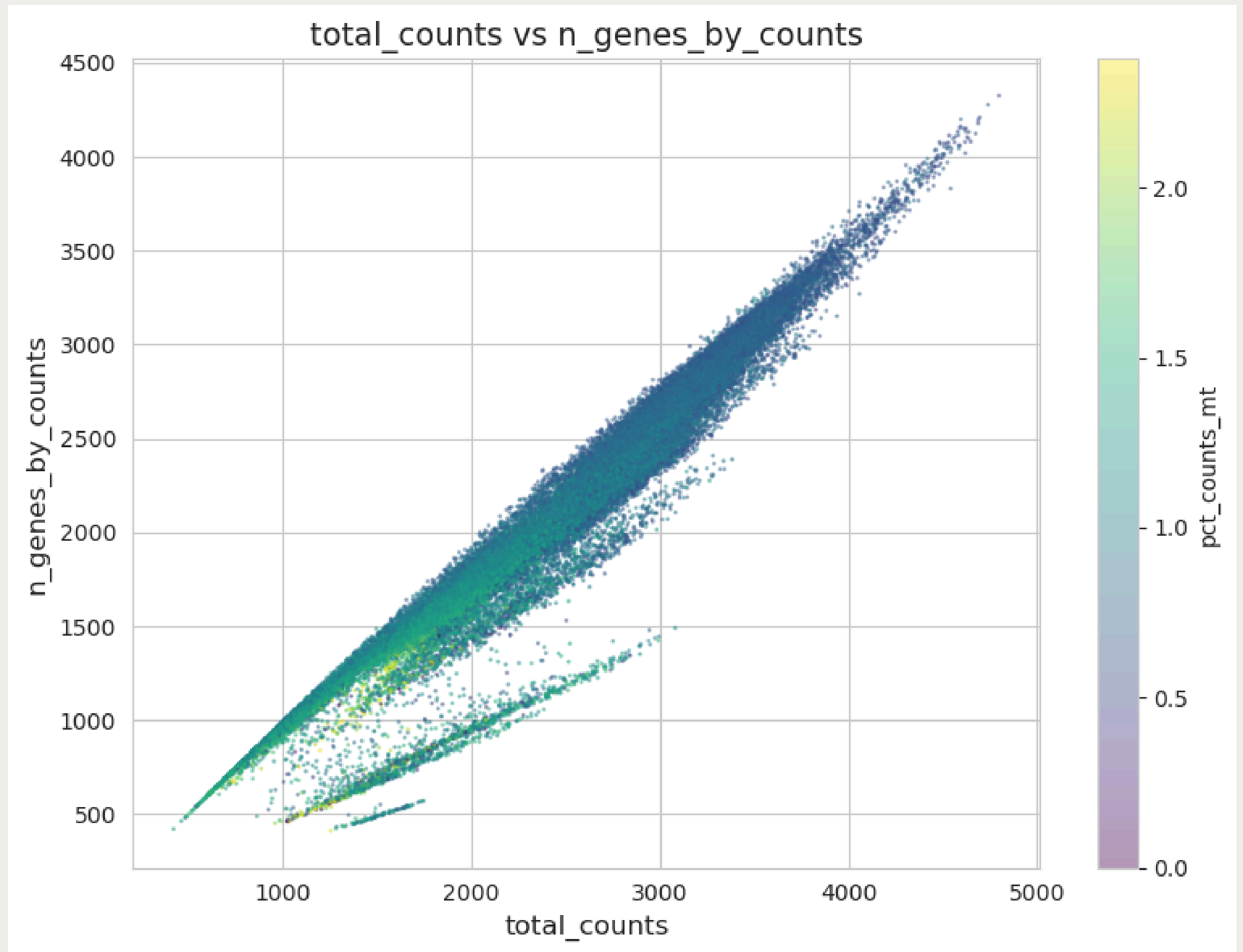
Gene detection is comparable across all 8 samples post-QC



After QC, MT% is low across all batches, well within acceptable range

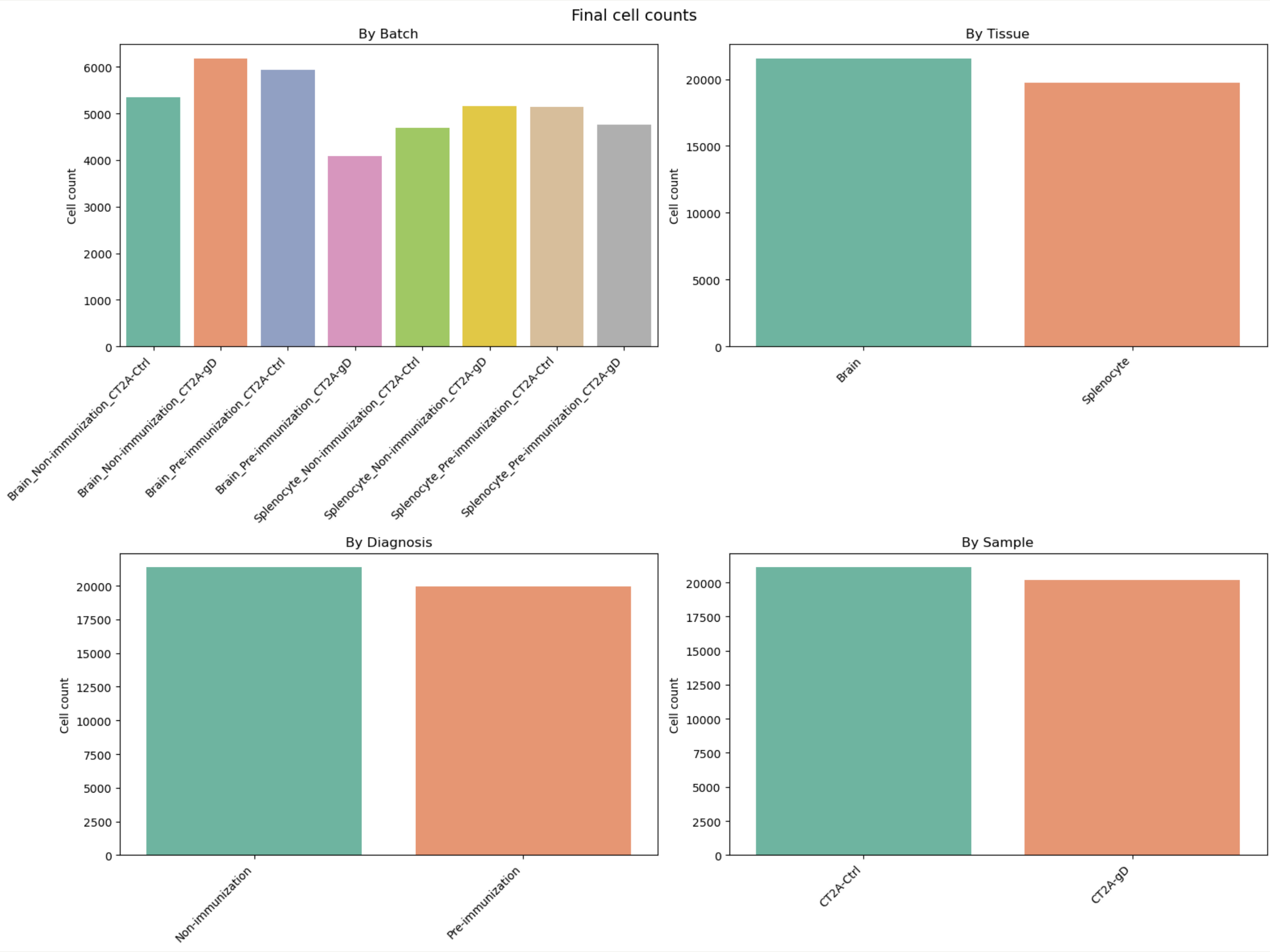


Strong linear
relationship
between UMI
counts and gene
detection,
consistent with
high-quality cells



Final cell counts

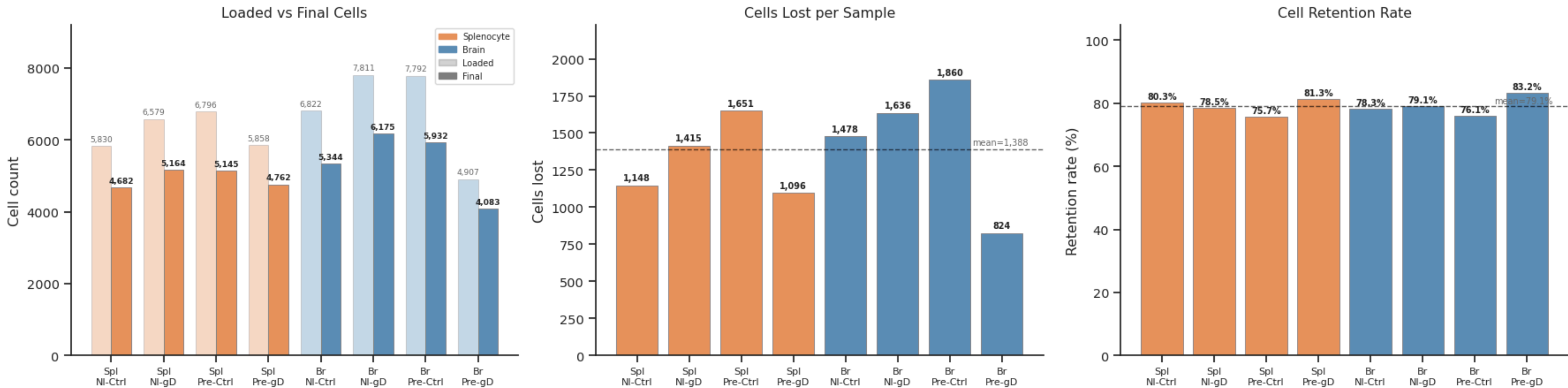
~41,000 cells retained after QC, roughly balanced across conditions and tissue types



Mean retention rate of ~79% across all samples

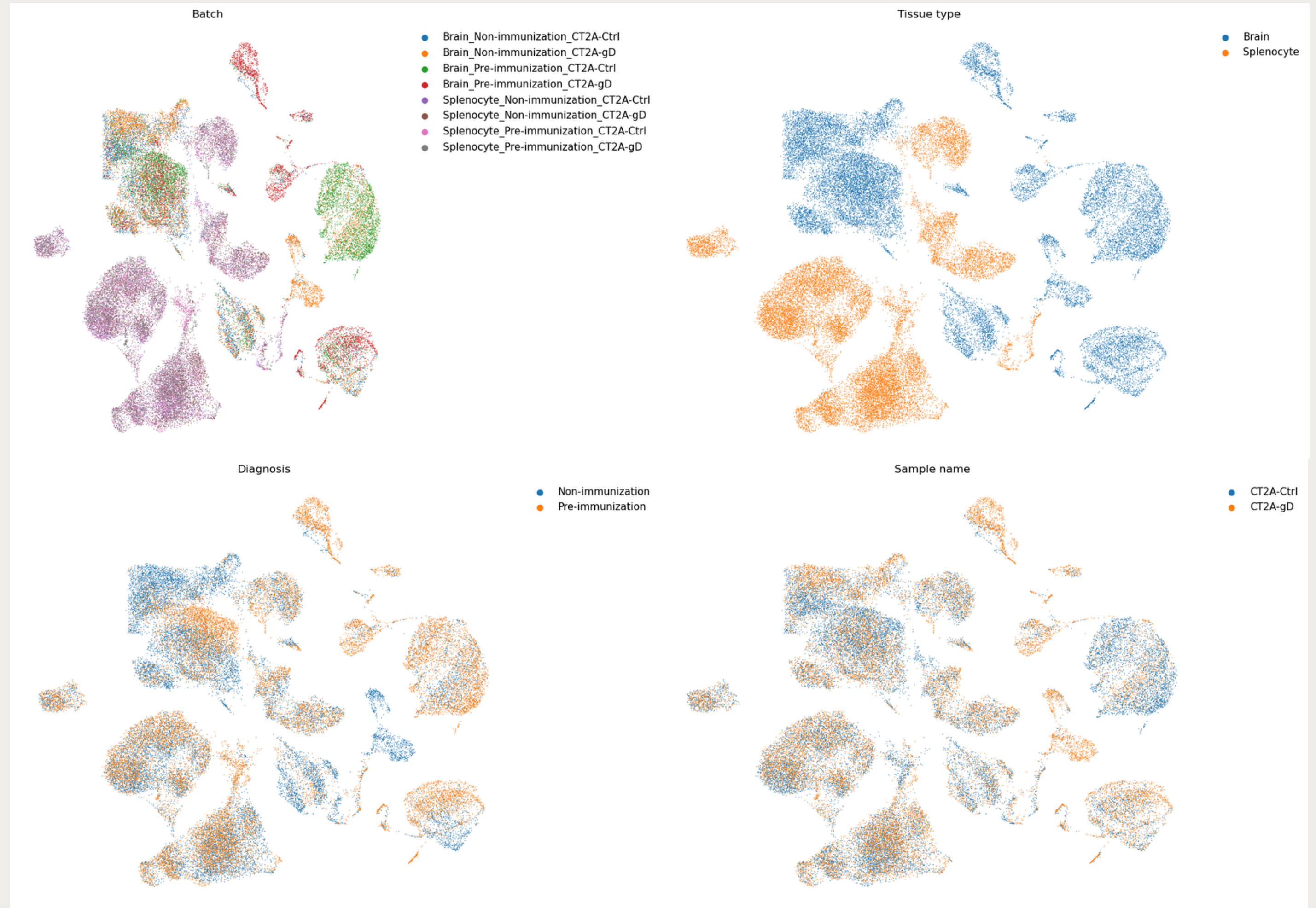
Sufficient cell numbers retained consistently across all batches for downstream analysis

Cell Count Summary Across QC Pipeline



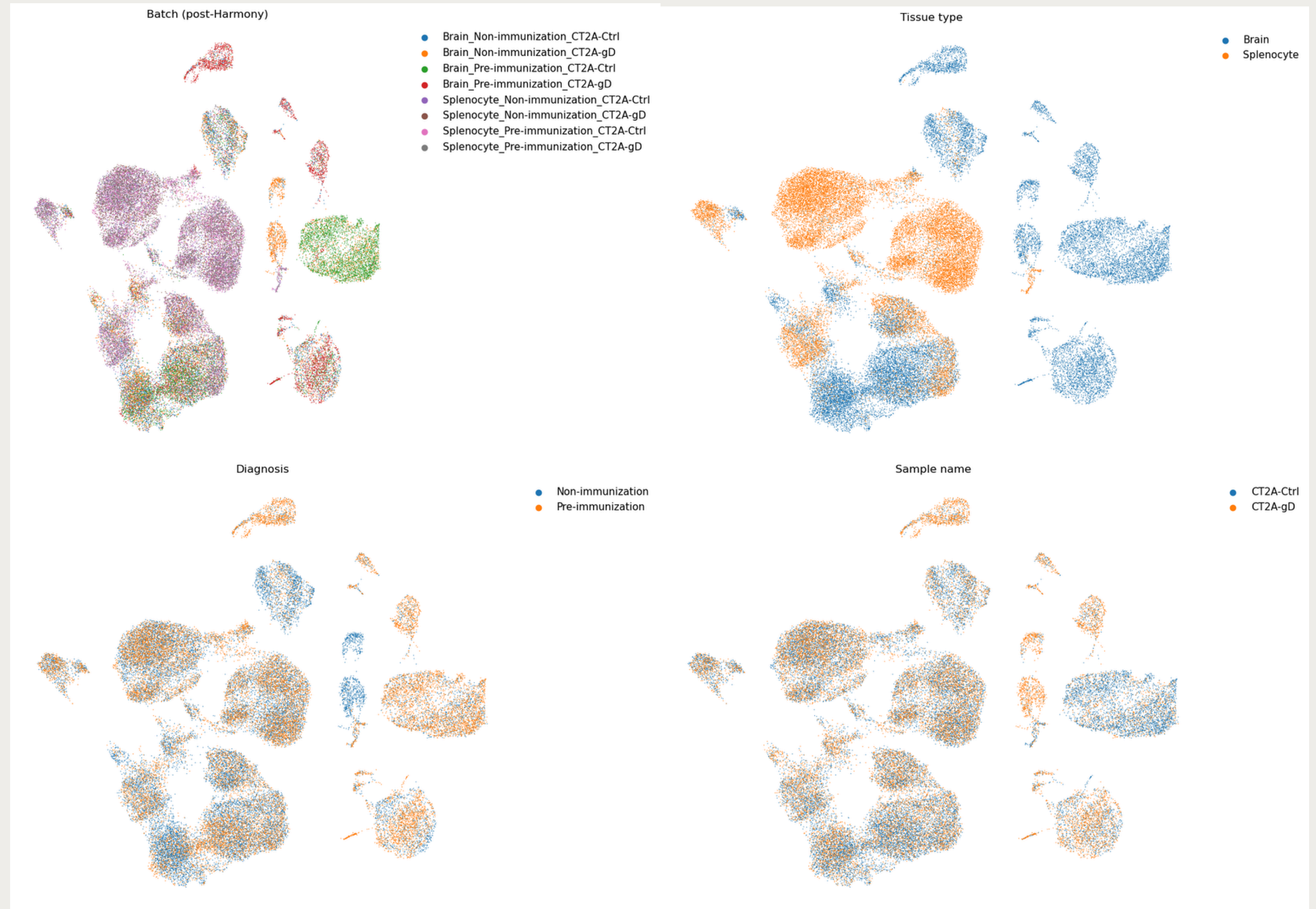
Integration

Pre-integration UMAP shows batch-driven clustering, with sample identity driving cluster structure across samples



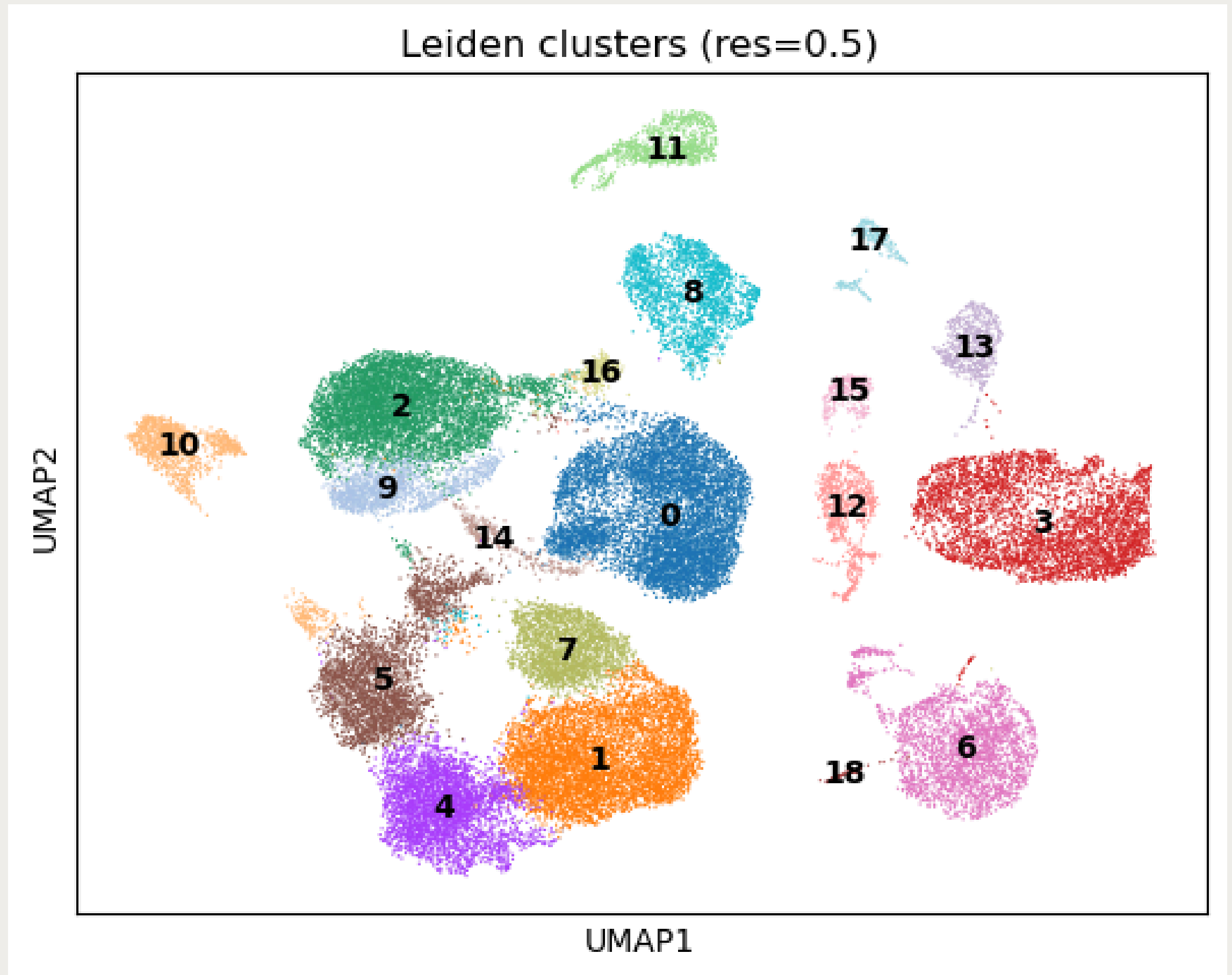
After Harmony

Post-Harmony UMAP shows improved sample mixing, with batch effects reduced and biological variation retained

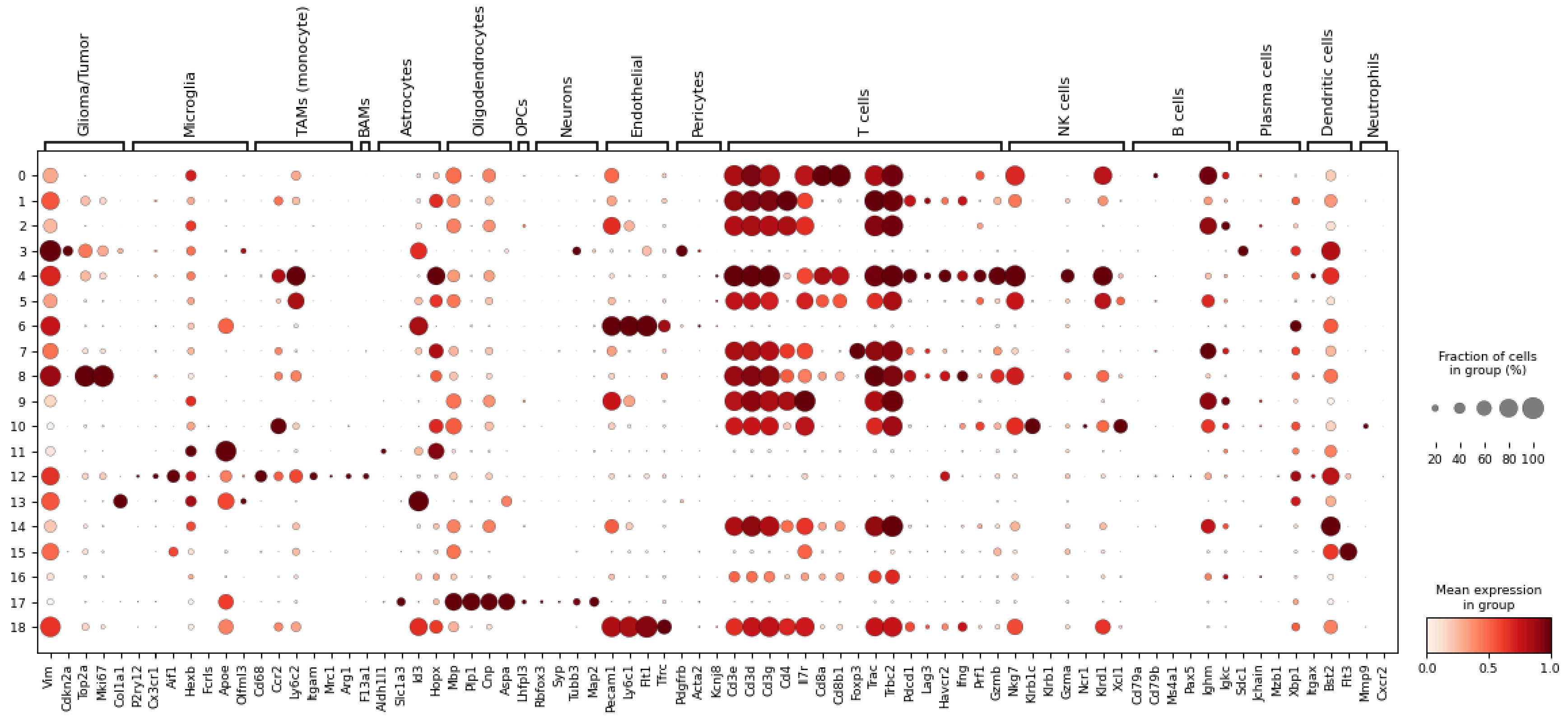


Clusters

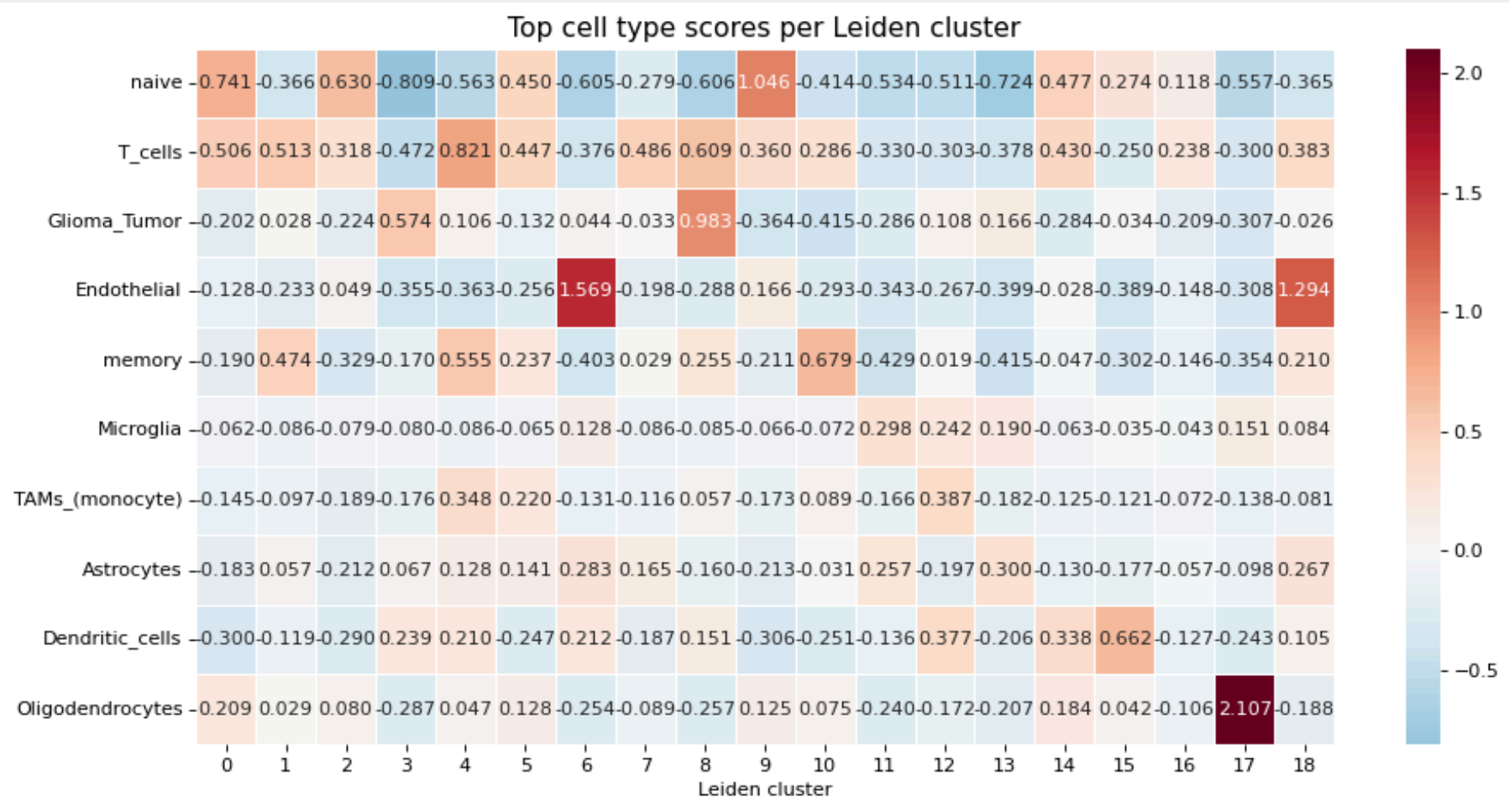
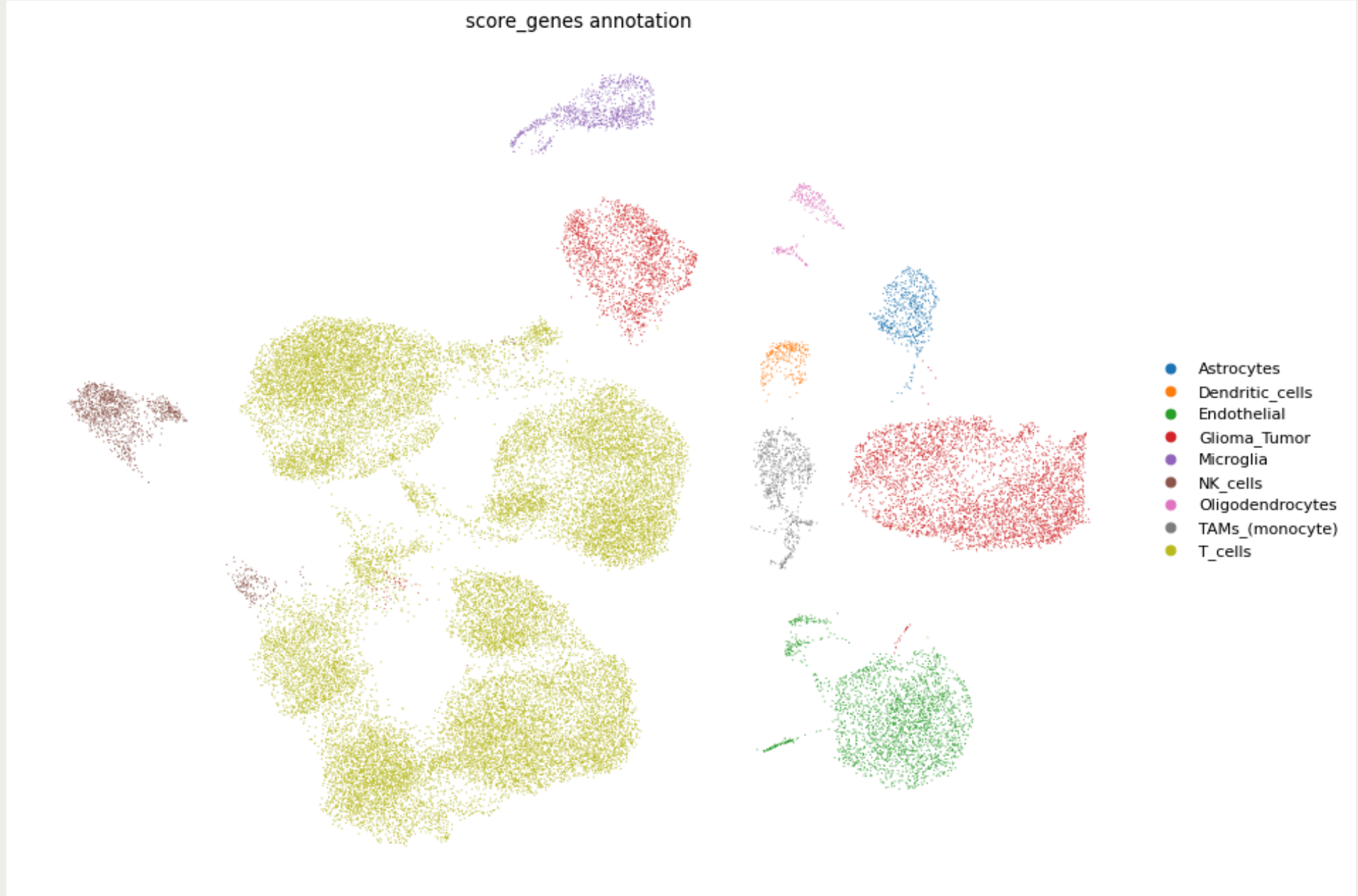
- Leiden clustering at resolution 0.5 yields 19 clusters with clear population separation



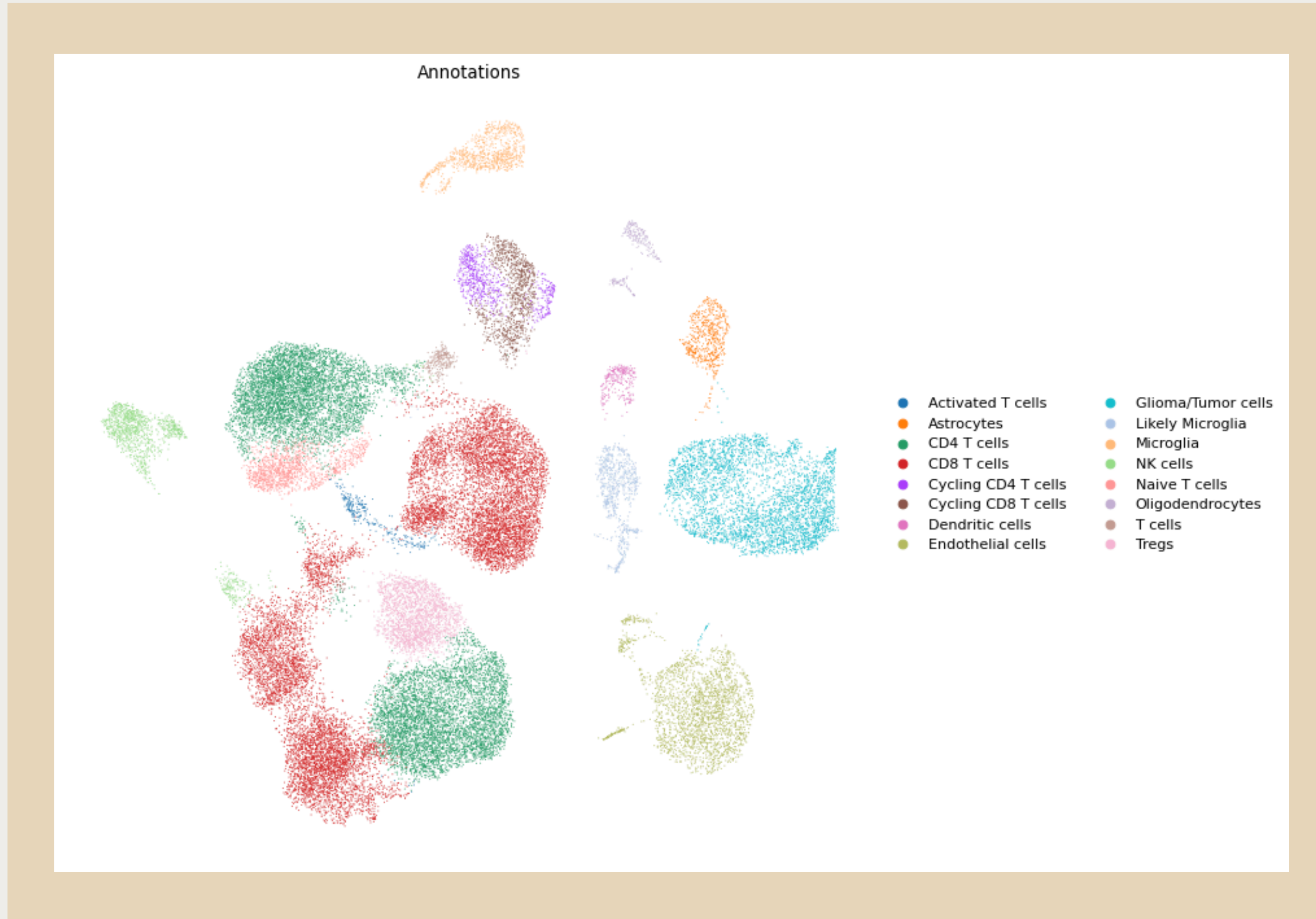
Marker gene expression supports broad cell type assignments across all clusters



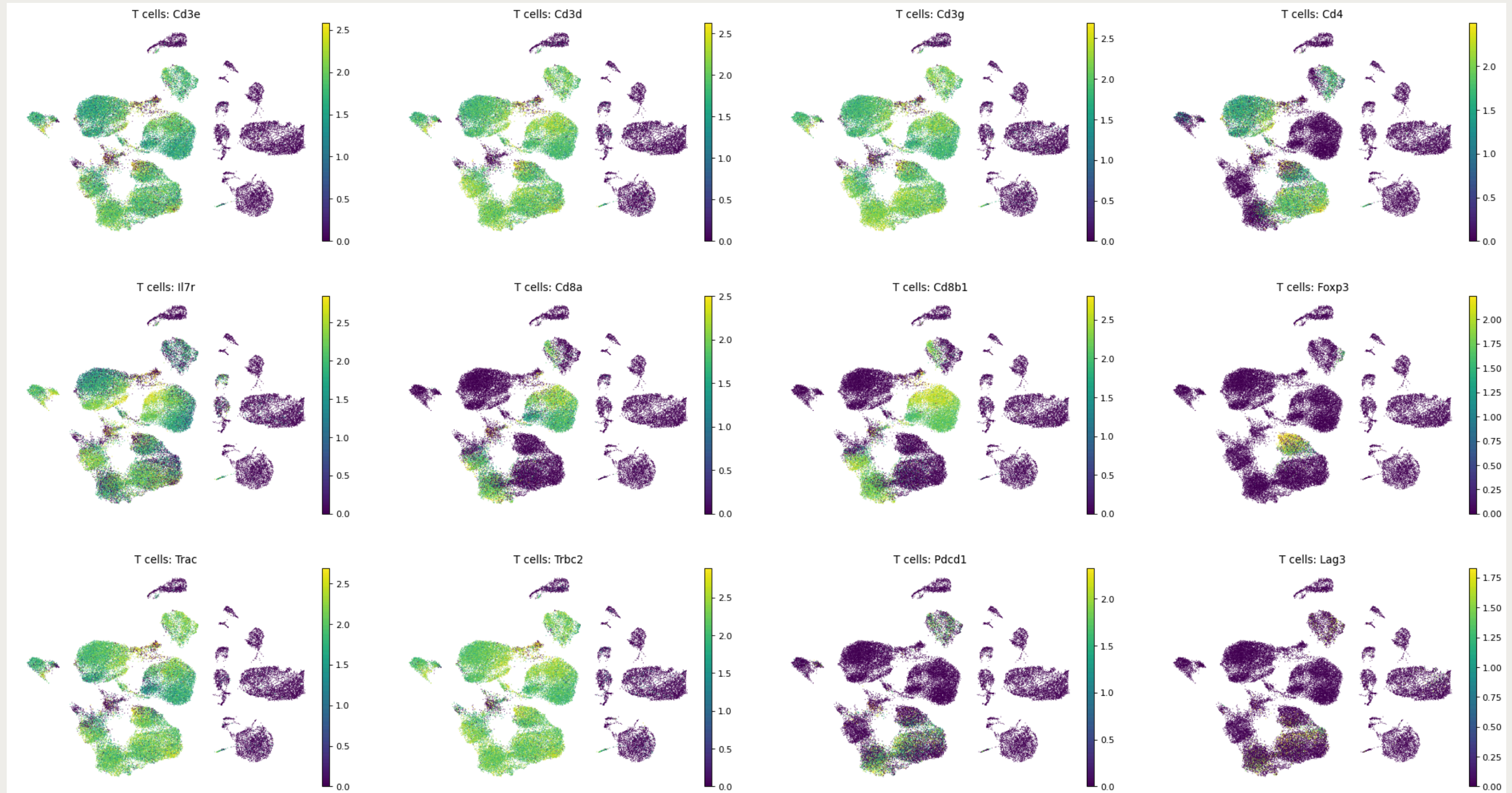
Gene module scores used to cross-reference and validate manual cluster annotations



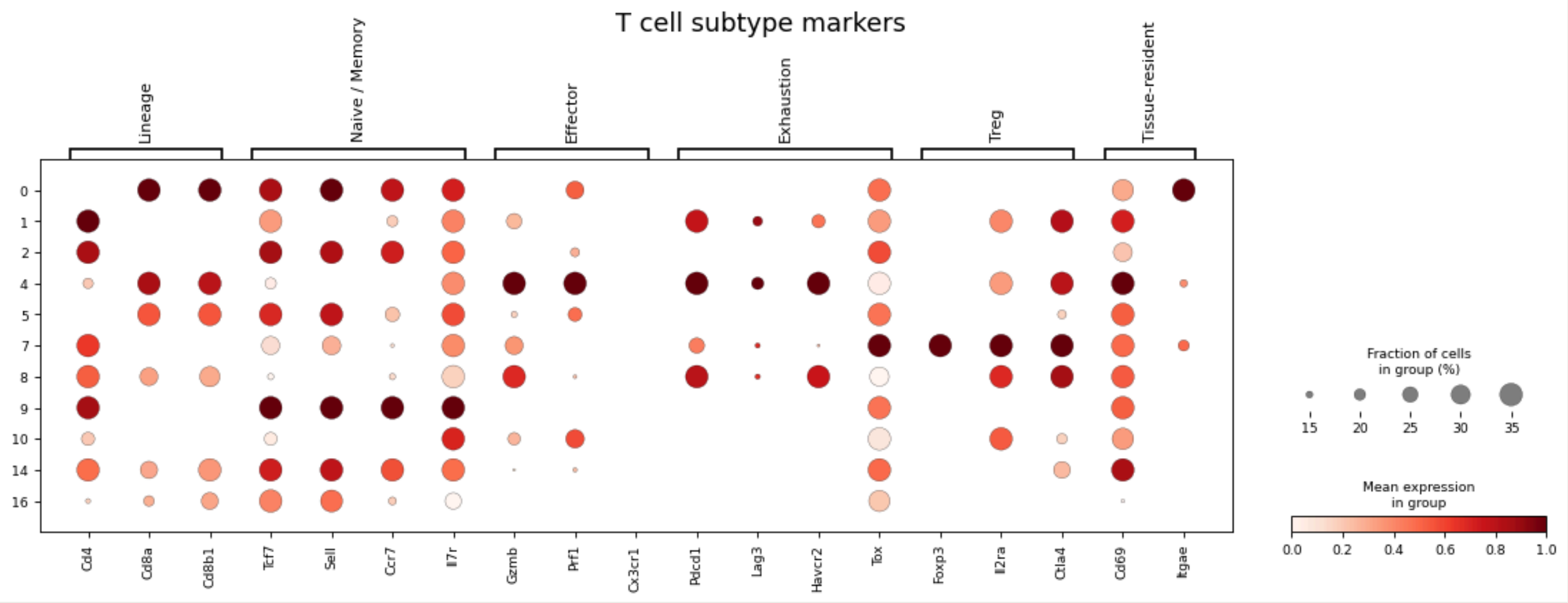
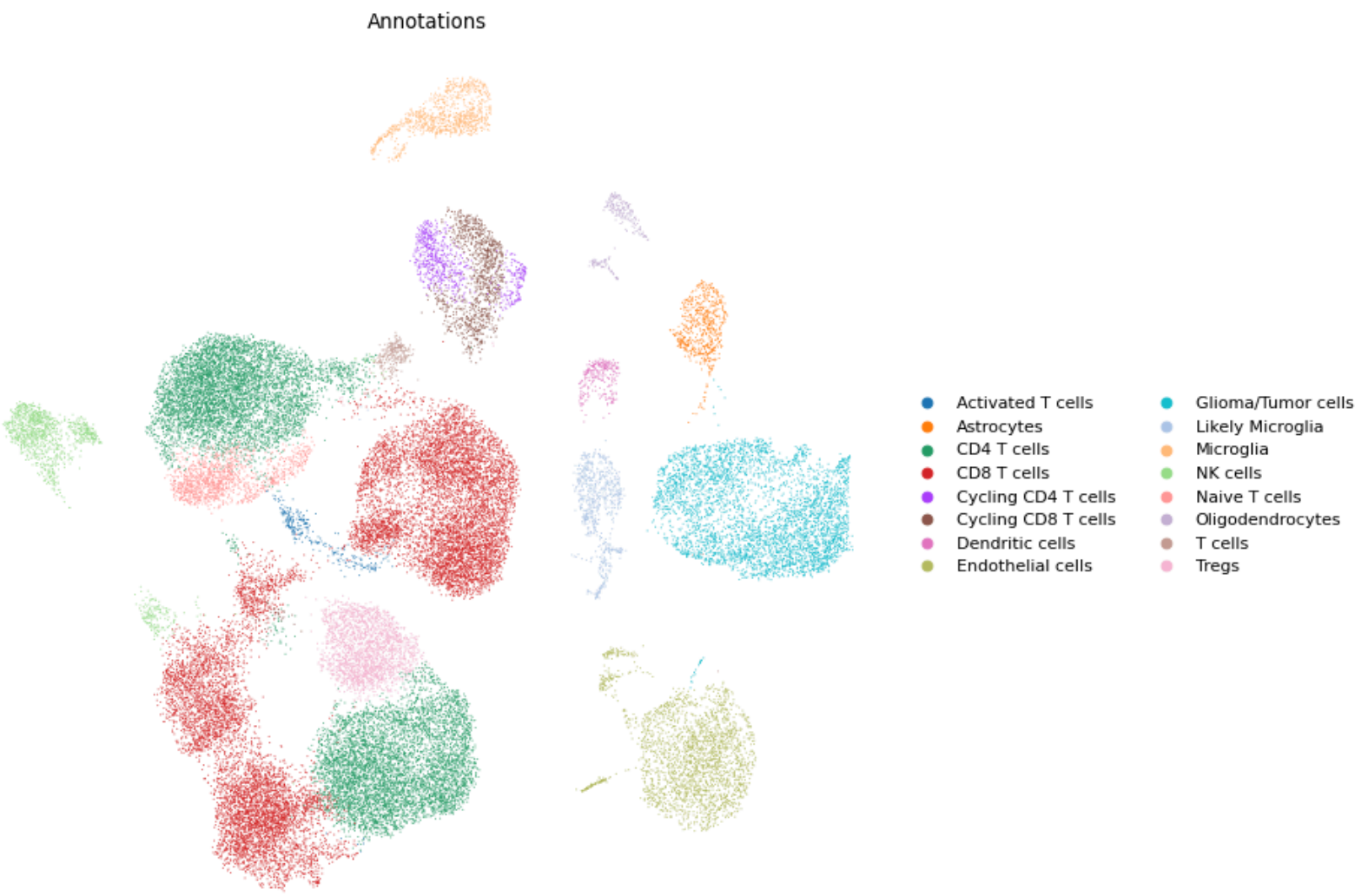
Integrated dataset resolved into distinct cell populations across brain and splenocyte compartments by manual annotation



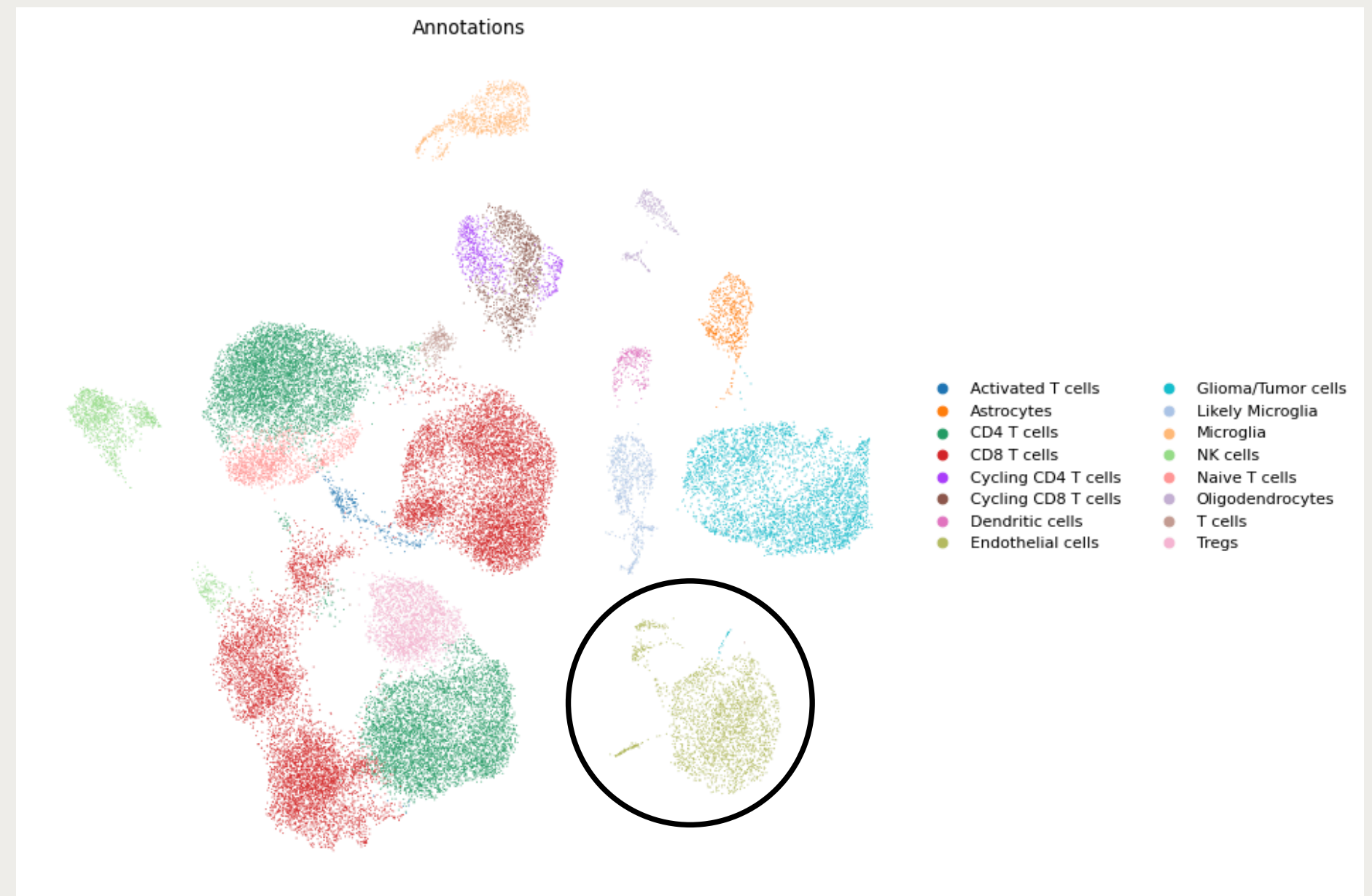
T cell subclusters identified by using markers



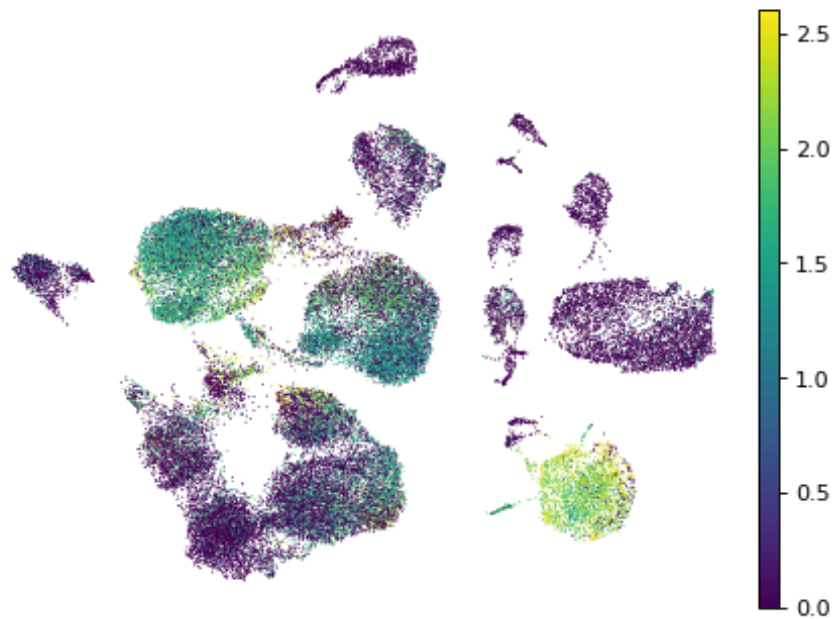
Dot plot and UMAP confirm distinct T cell subpopulations in clusters



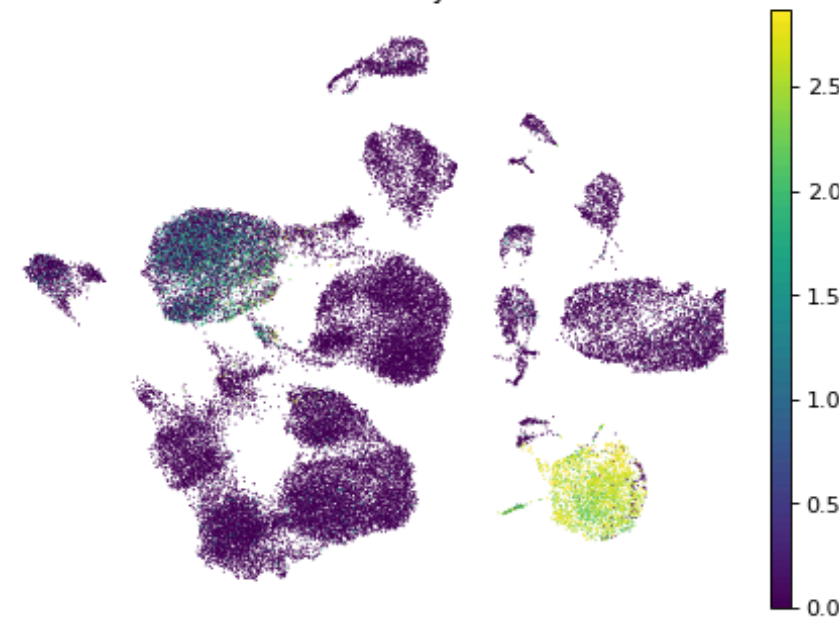
Endothelial cluster
confirmed by
co-expression of
Pecam1, Ly6c1, Flt1,
and Tfrc



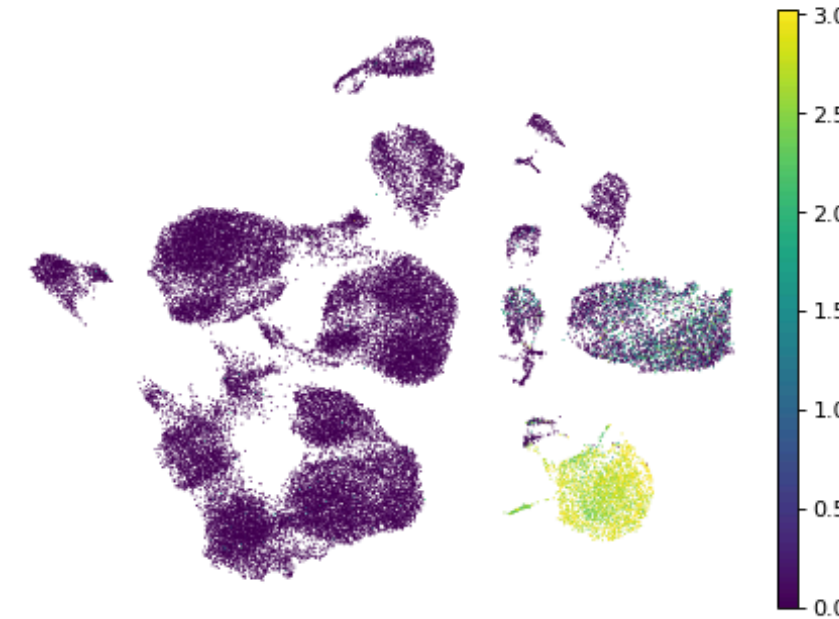
Endothelial: Pecam1



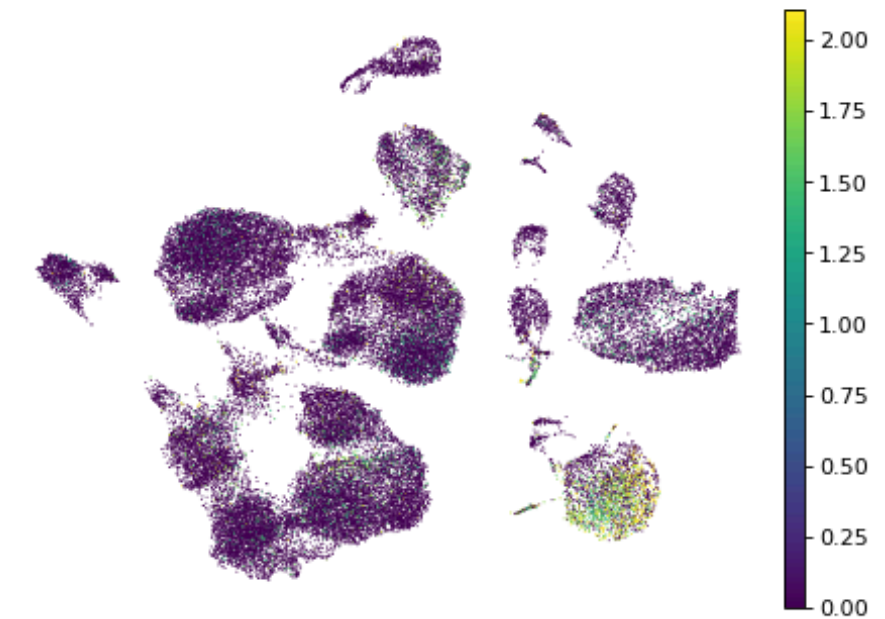
Endothelial: Ly6c1



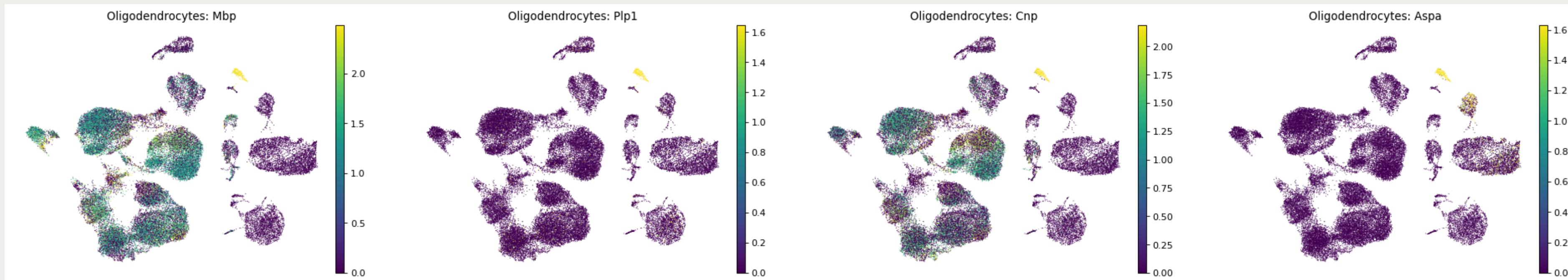
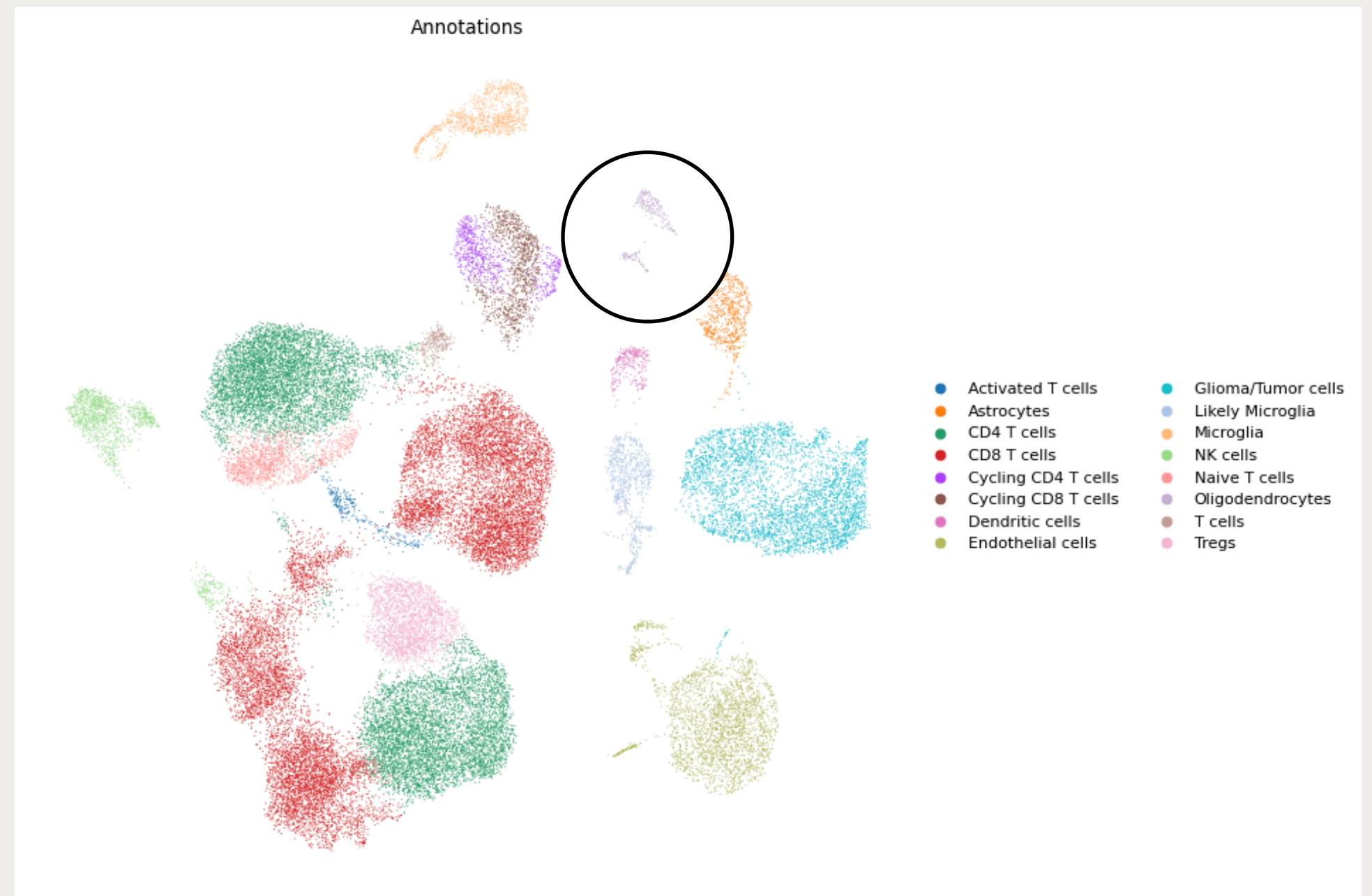
Endothelial: Flt1



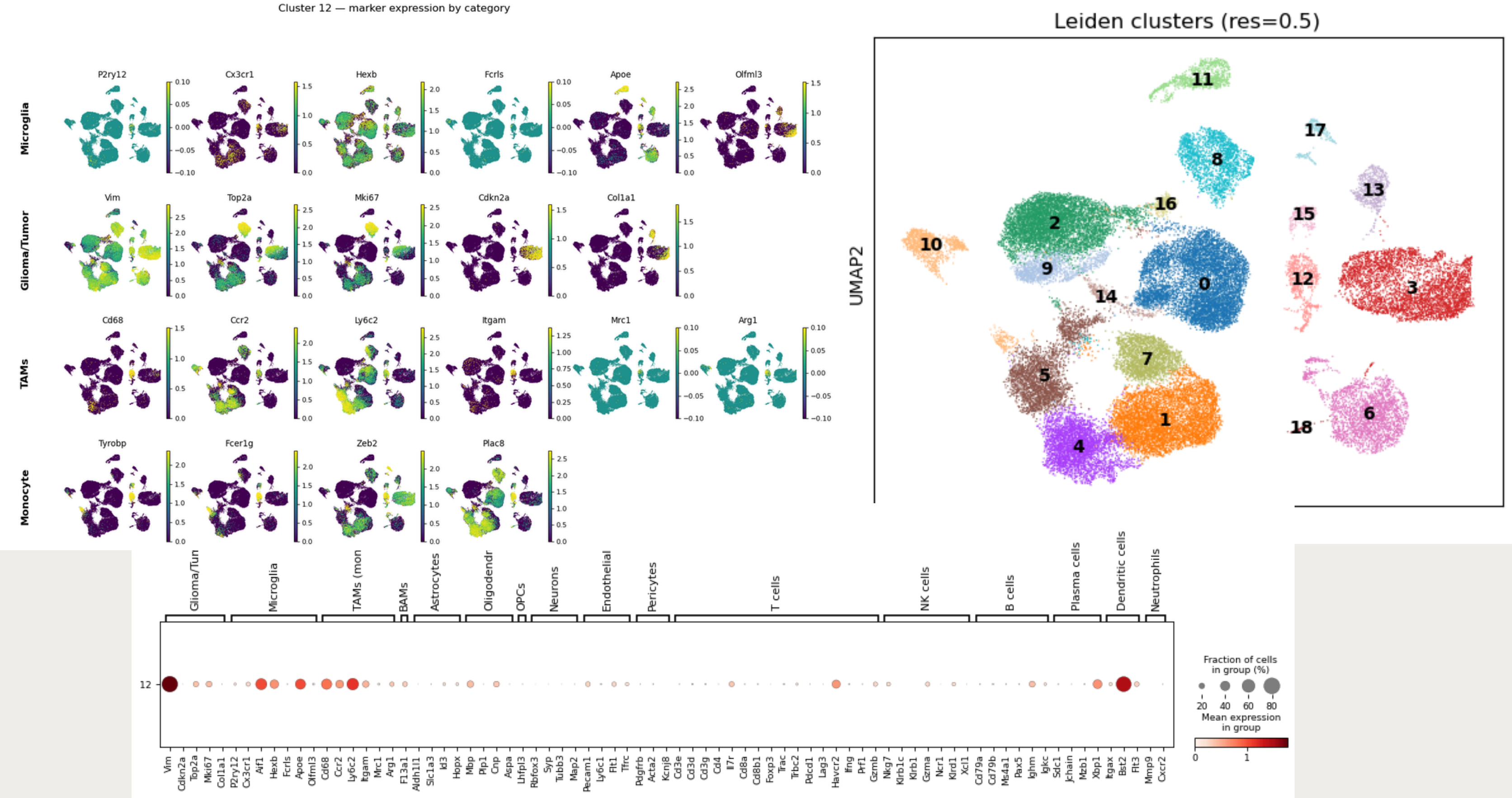
Endothelial: Tfrc



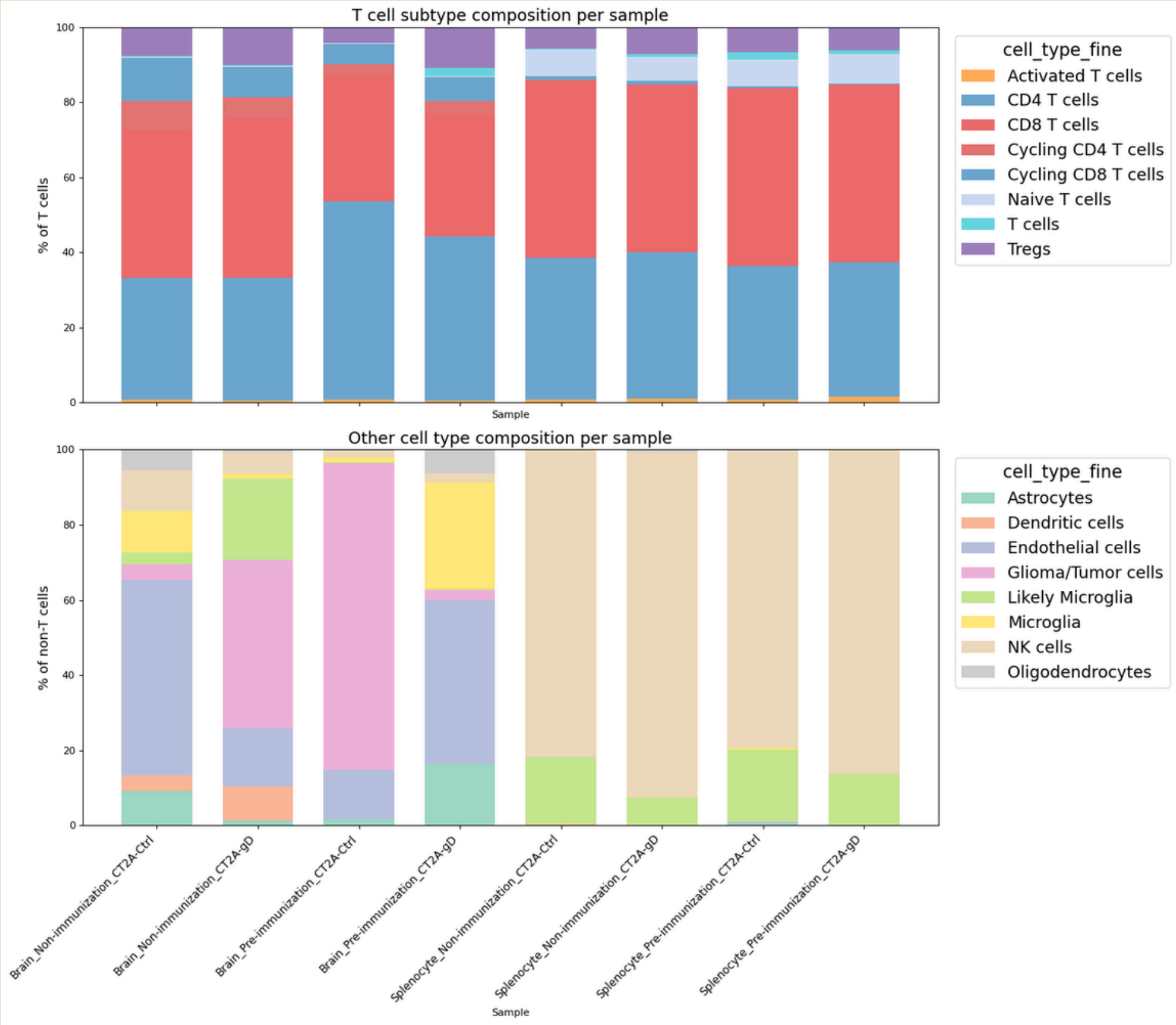
Oligodendrocyte cluster confirmed by co-expression of Mbp, Plp1, Cnp, and Aspa



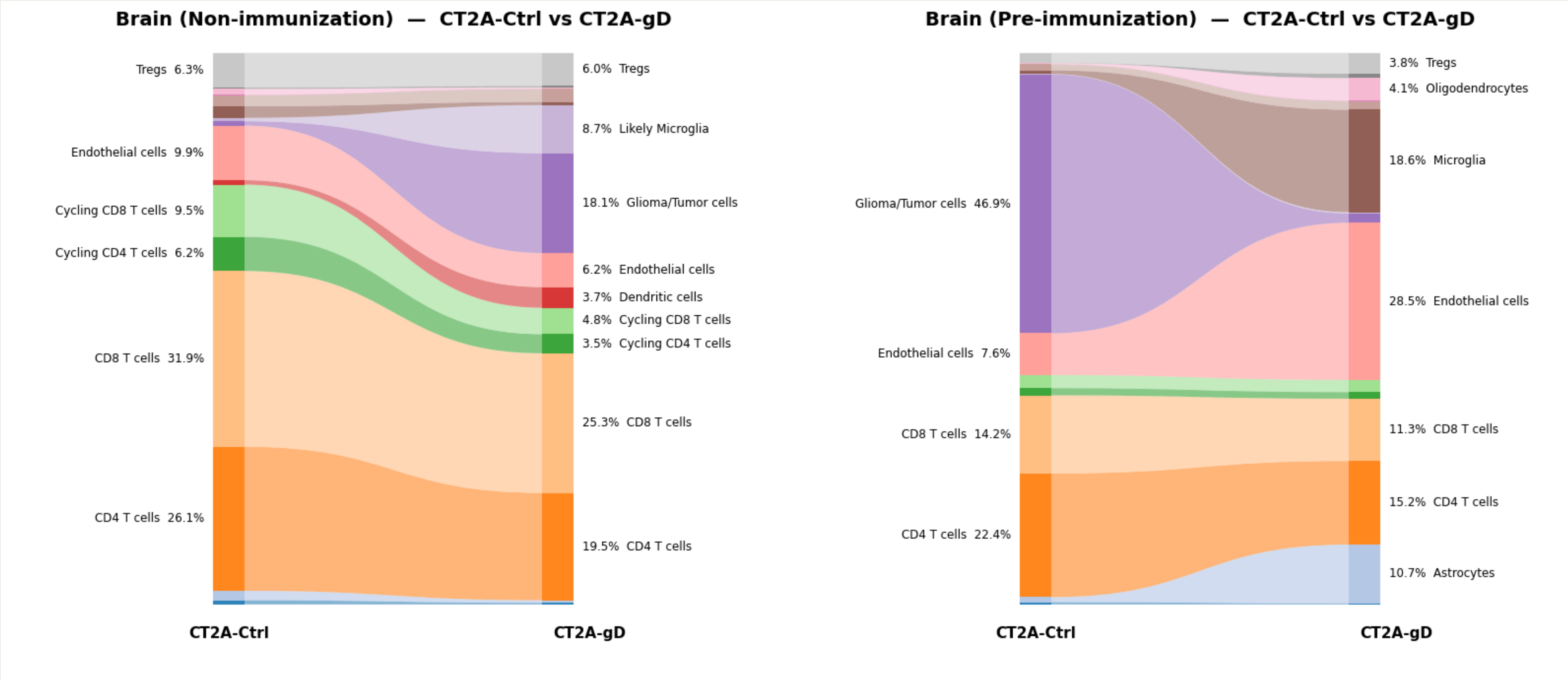
Cluster 12 identity is unclear — marker expression is inconsistent with a single cell type. Currently labeled as “Likely Microglia”



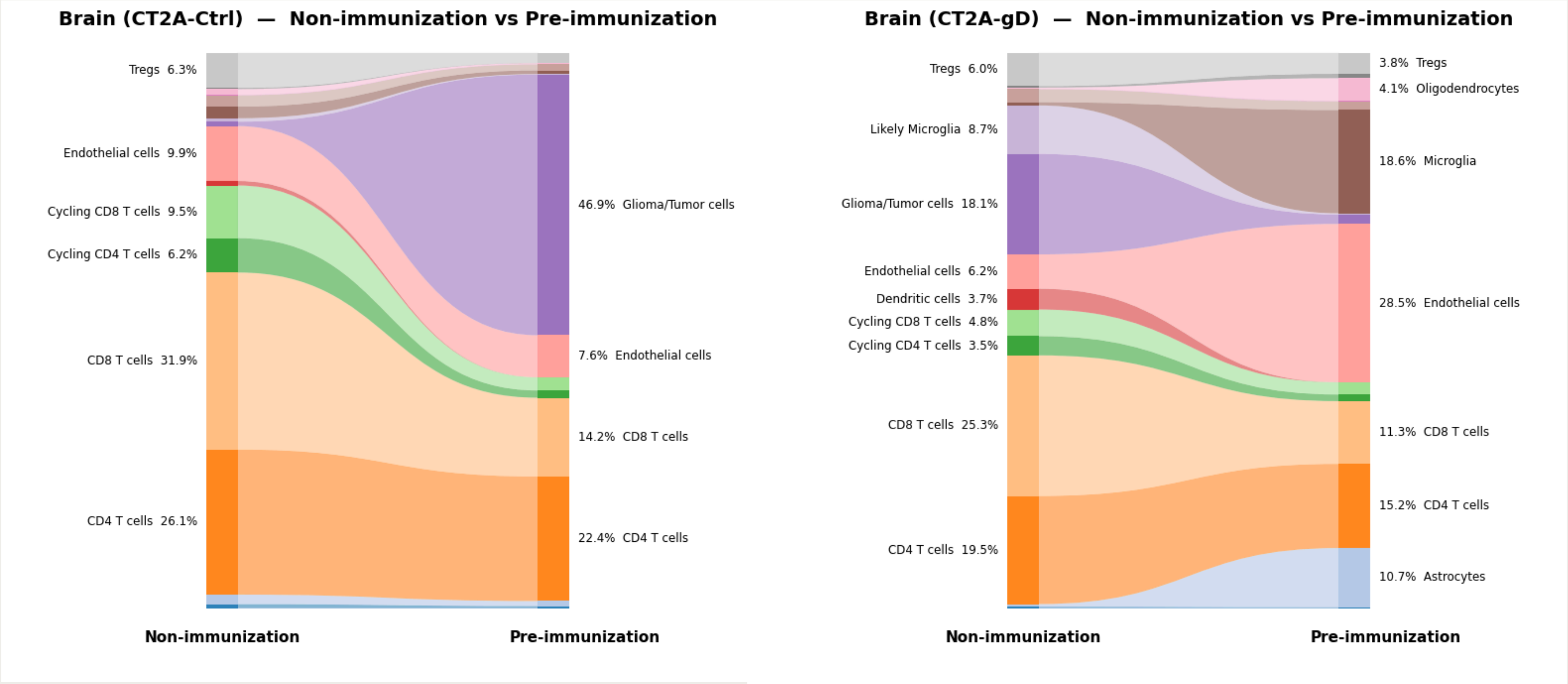
Cell type proportions vary across samples, split between T cells and non-T cells



Brain immunization comparison, split by CT2A-Ctrl and CT2A-gD

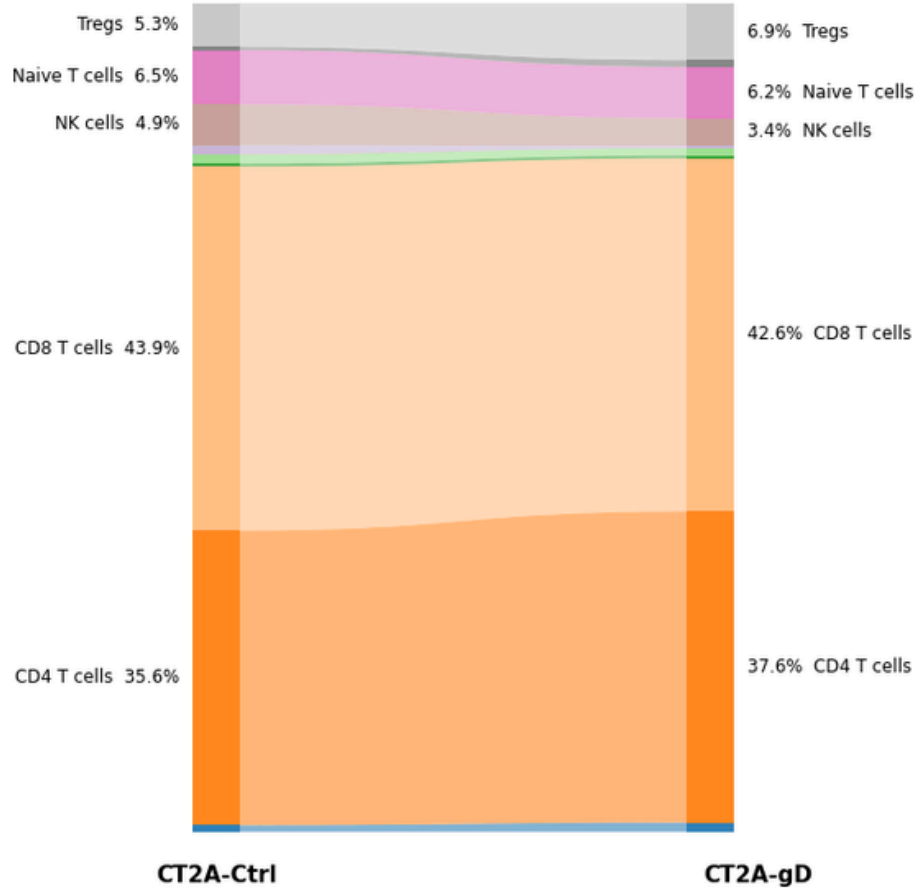


Brain tumor-type comparison, split by non- and pre-immunization

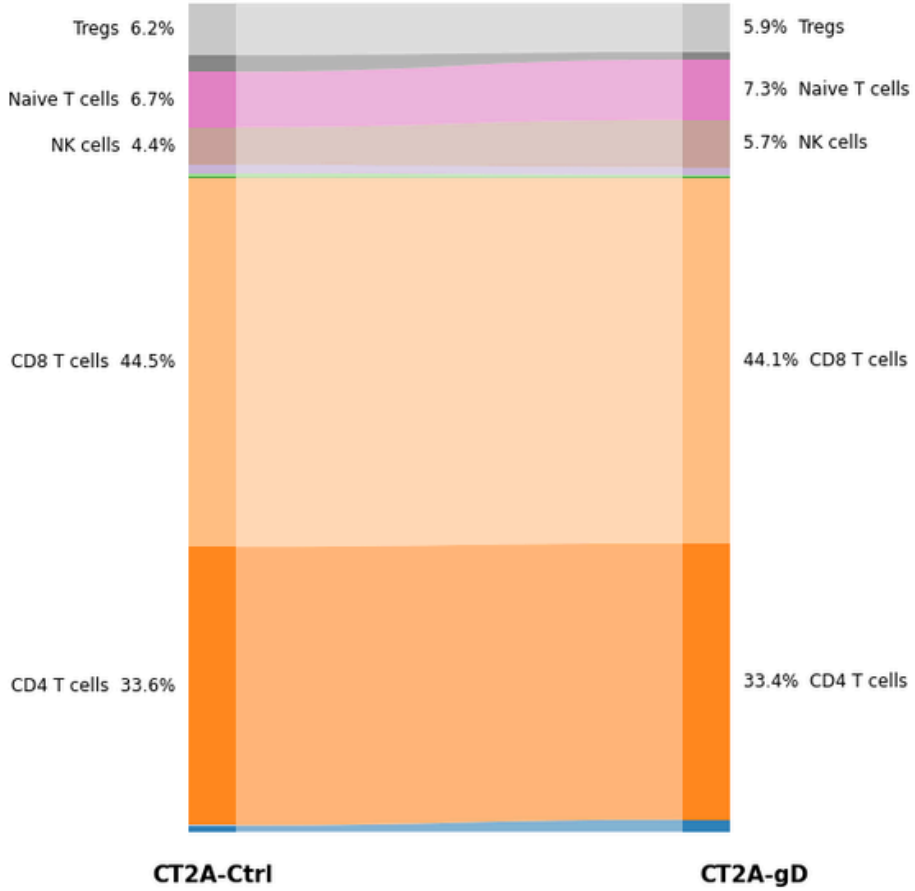


Splenocyte
composition
is largely
stable across
all conditions.

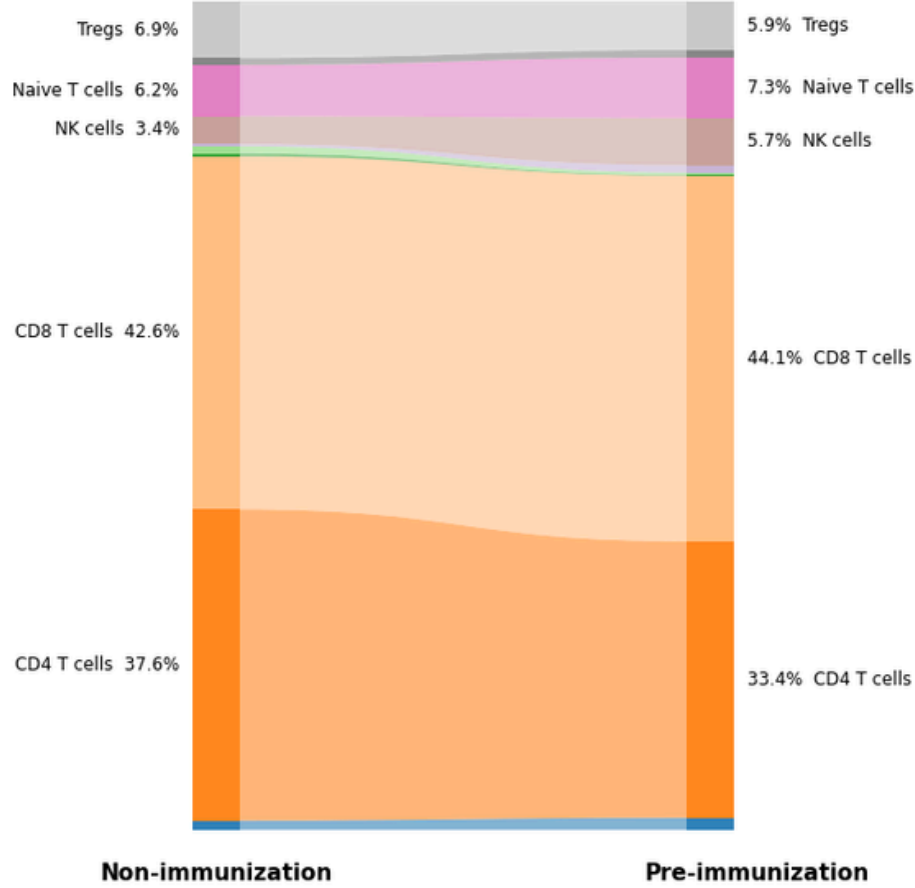
Splenocyte (Non-immunization) — CT2A-Ctrl vs CT2A-gD



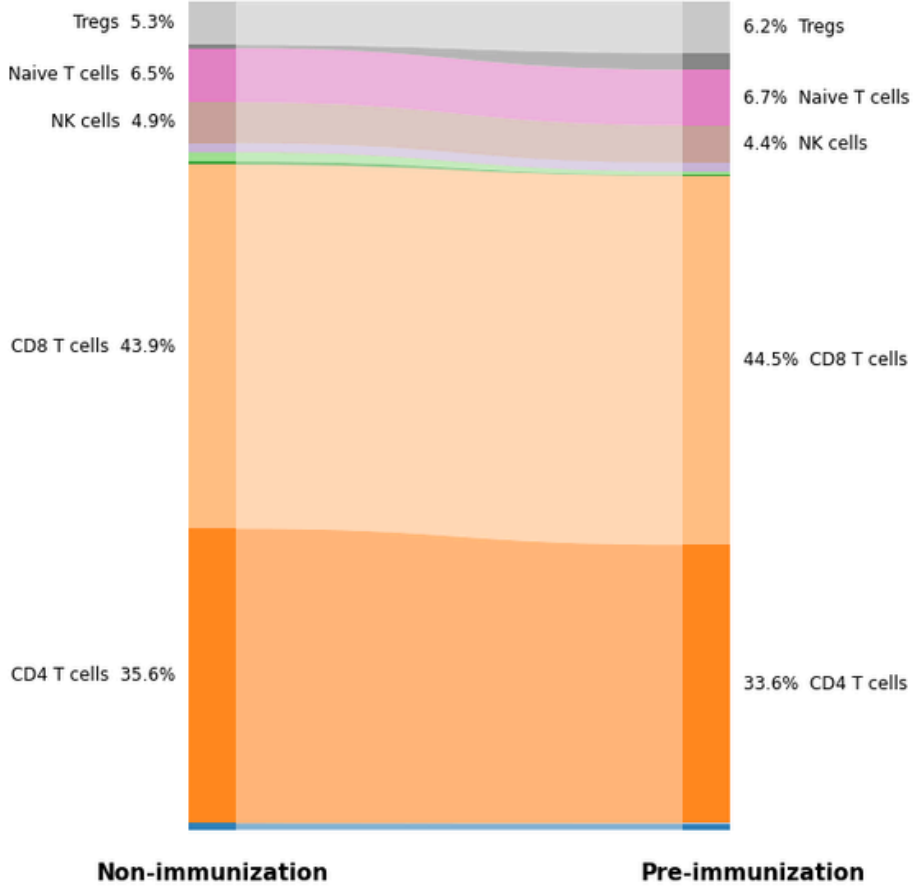
Splenocyte (Pre-immunization) — CT2A-Ctrl vs CT2A-gD



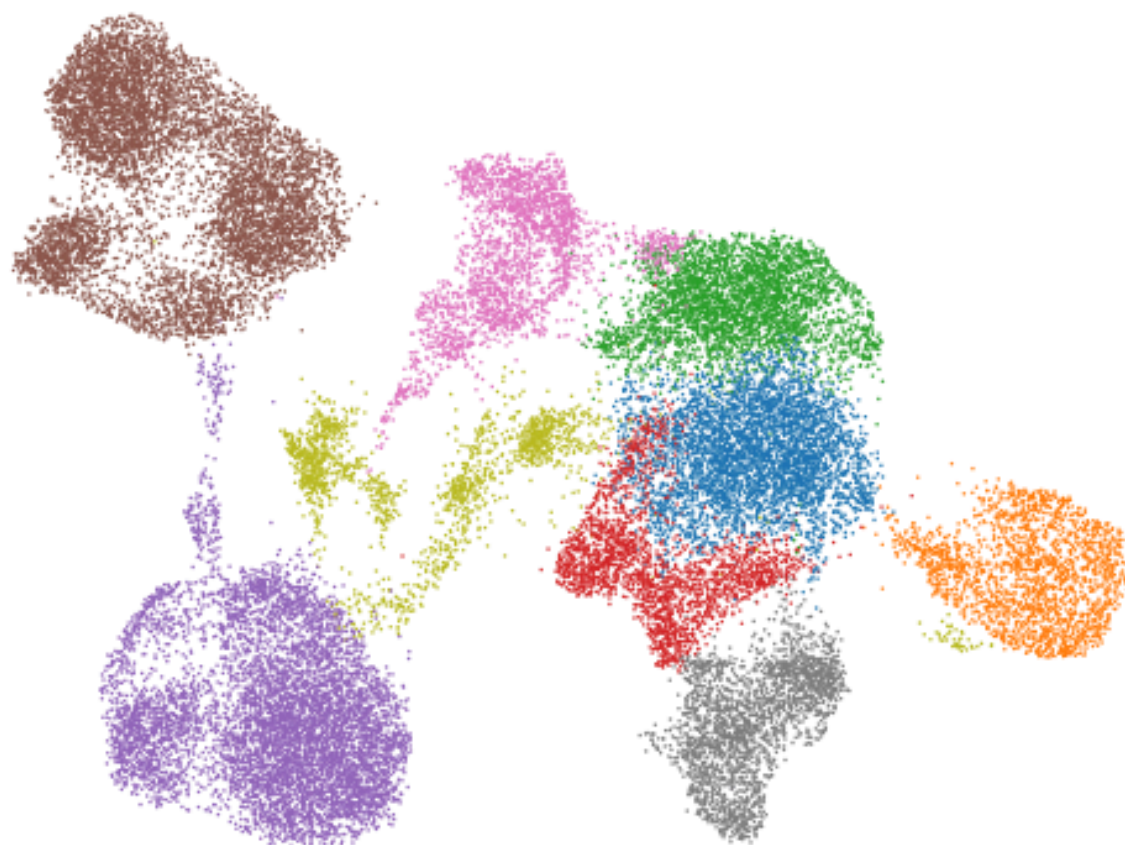
Splenocyte (CT2A-gD) — Non-immunization vs Pre-immunization



Splenocyte (CT2A-Ctrl) — Non-immunization vs Pre-immunization

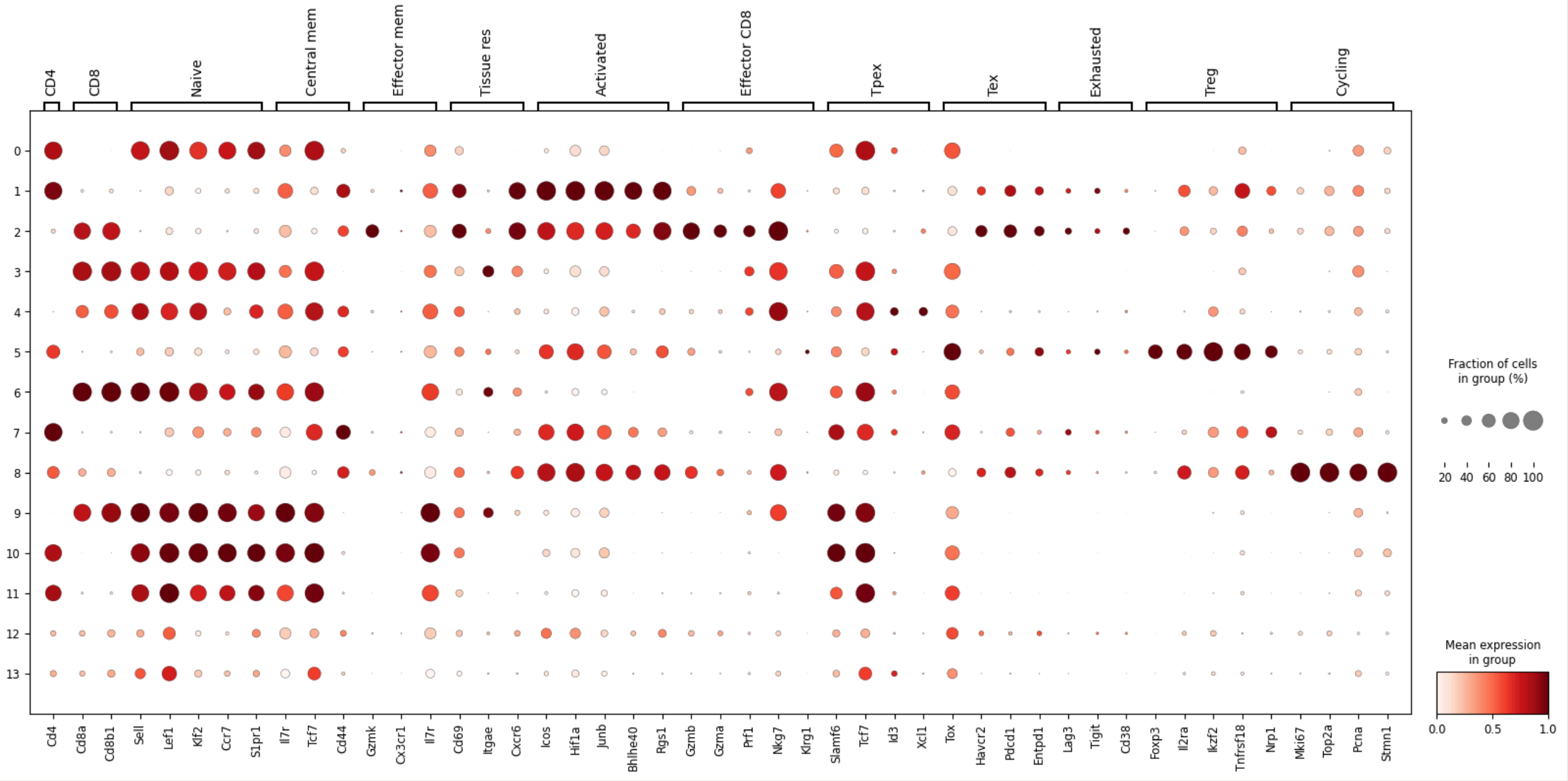


cell_type_tcell



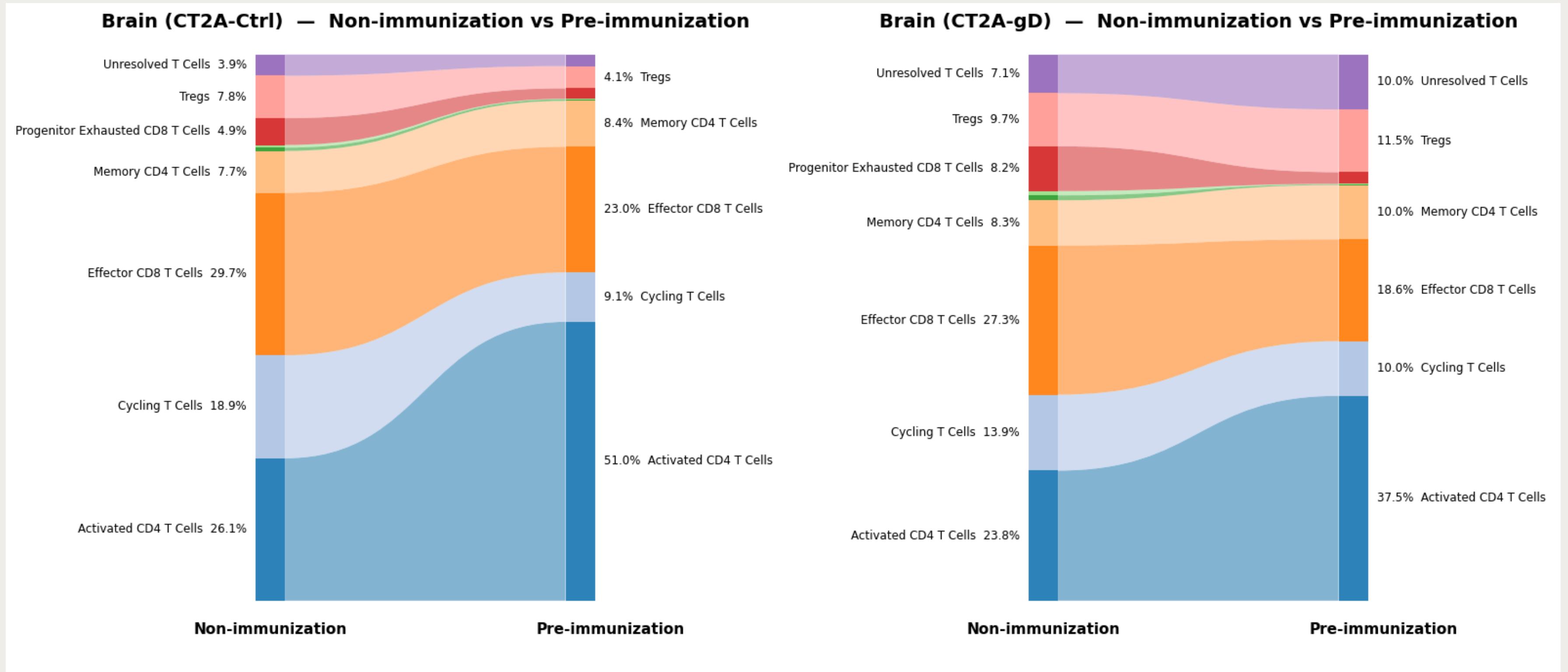
- Activated CD4 T Cells
- Cycling T Cells
- Effector CD8 T Cells
- Memory CD4 T Cells
- Naive CD4 T Cells
- Naive CD8 T Cells
- Progenitor Exhausted CD8 T Cells
- Tregs
- Unresolved T Cells

Subclustering T cells Further



T cell subclustering

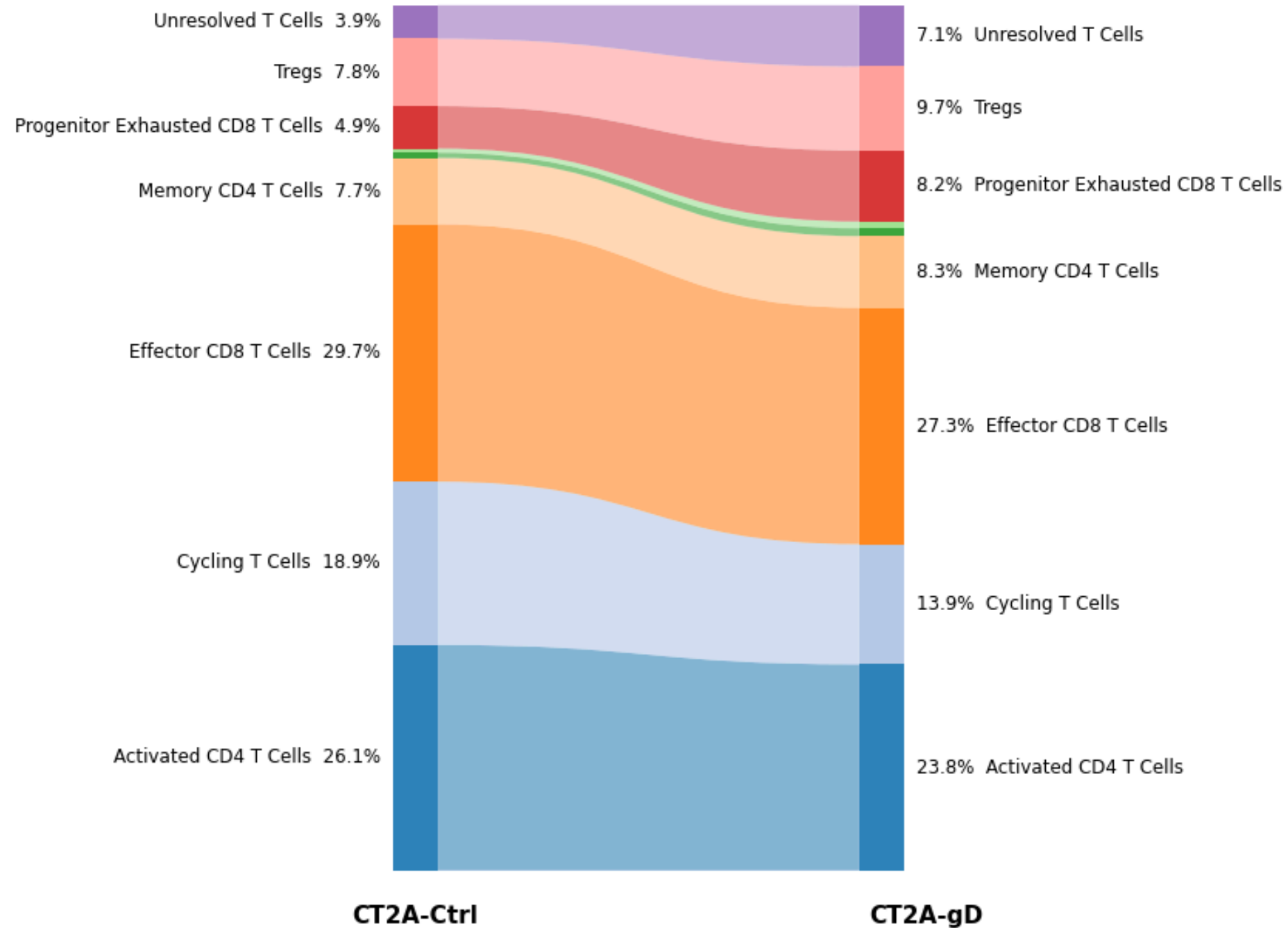
Brain immunization comparison, split by CT2A-Ctrl and CT2A-gD



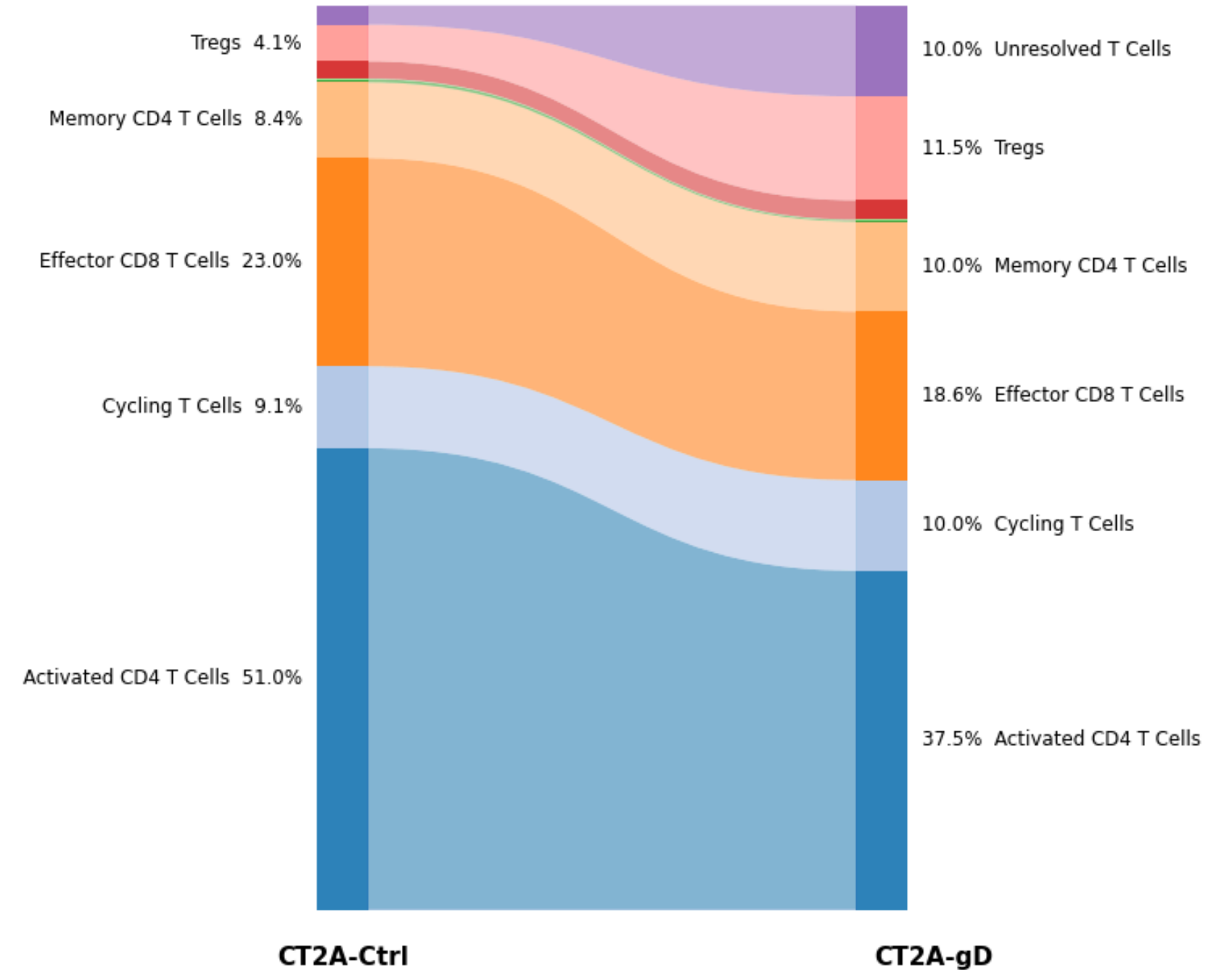
T cell subclustering

Brain tumor-type comparison, split by non- and pre-immunization

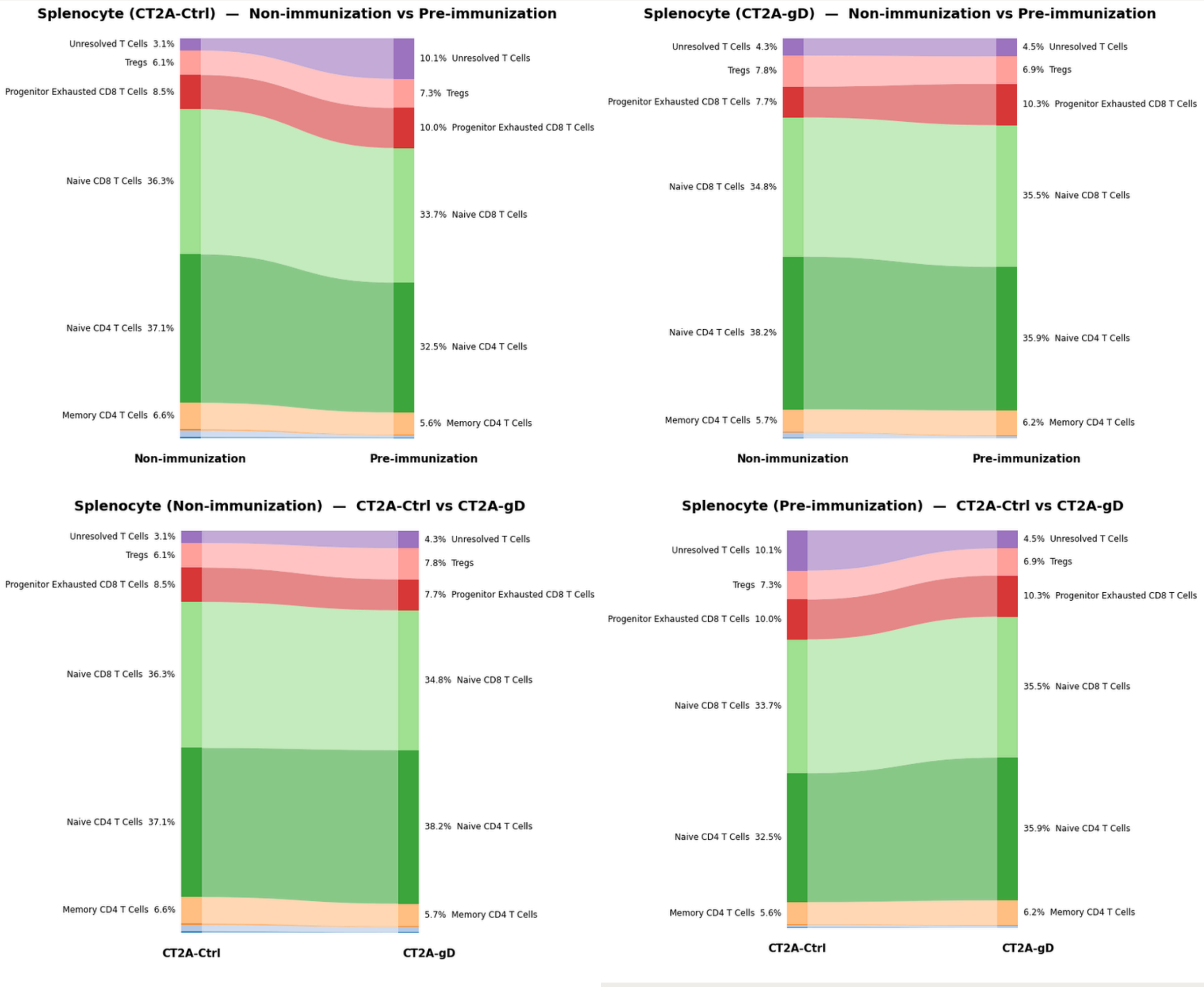
Brain (Non-immunization) — CT2A-Ctrl vs CT2A-gD



Brain (Pre-immunization) — CT2A-Ctrl vs CT2A-gD



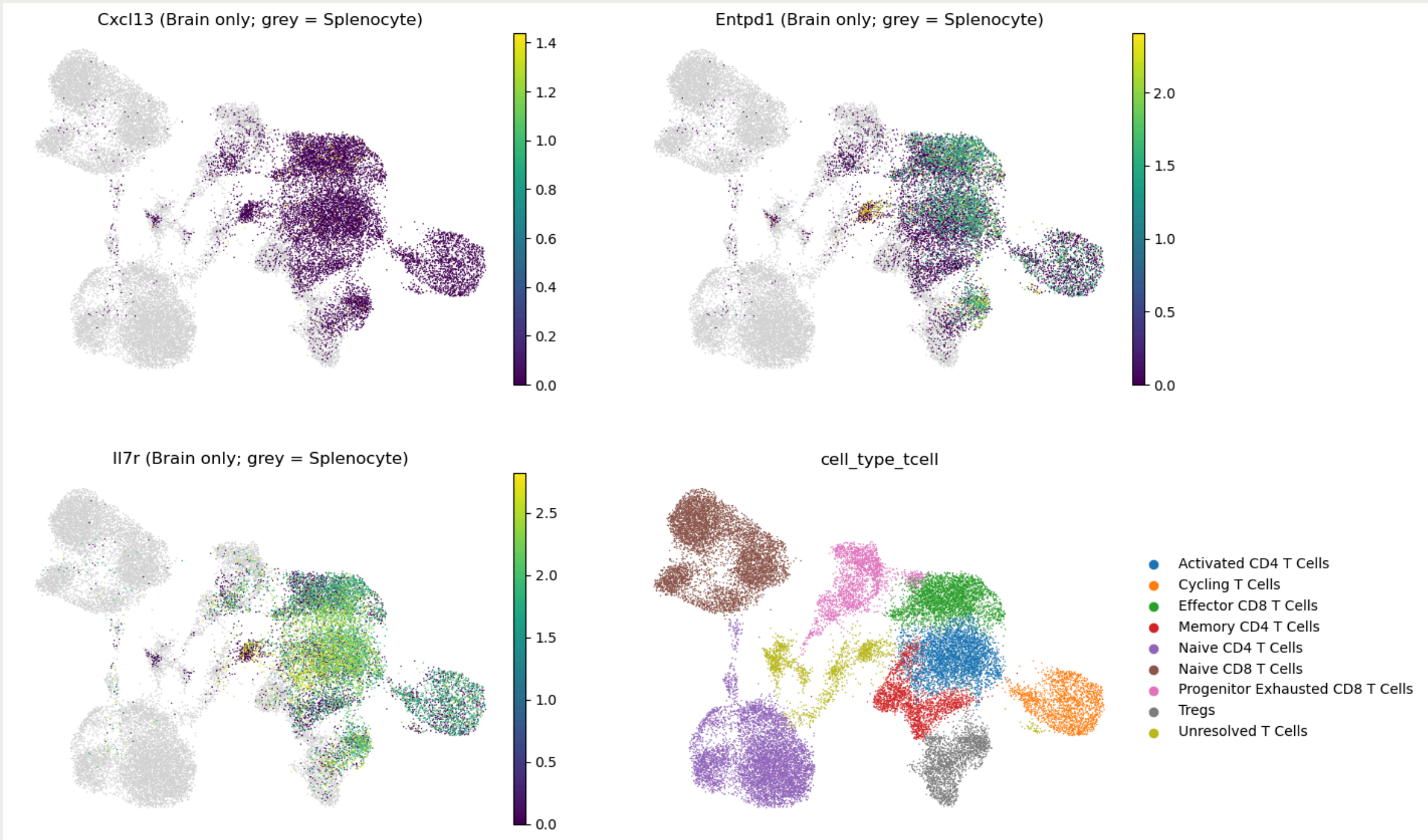
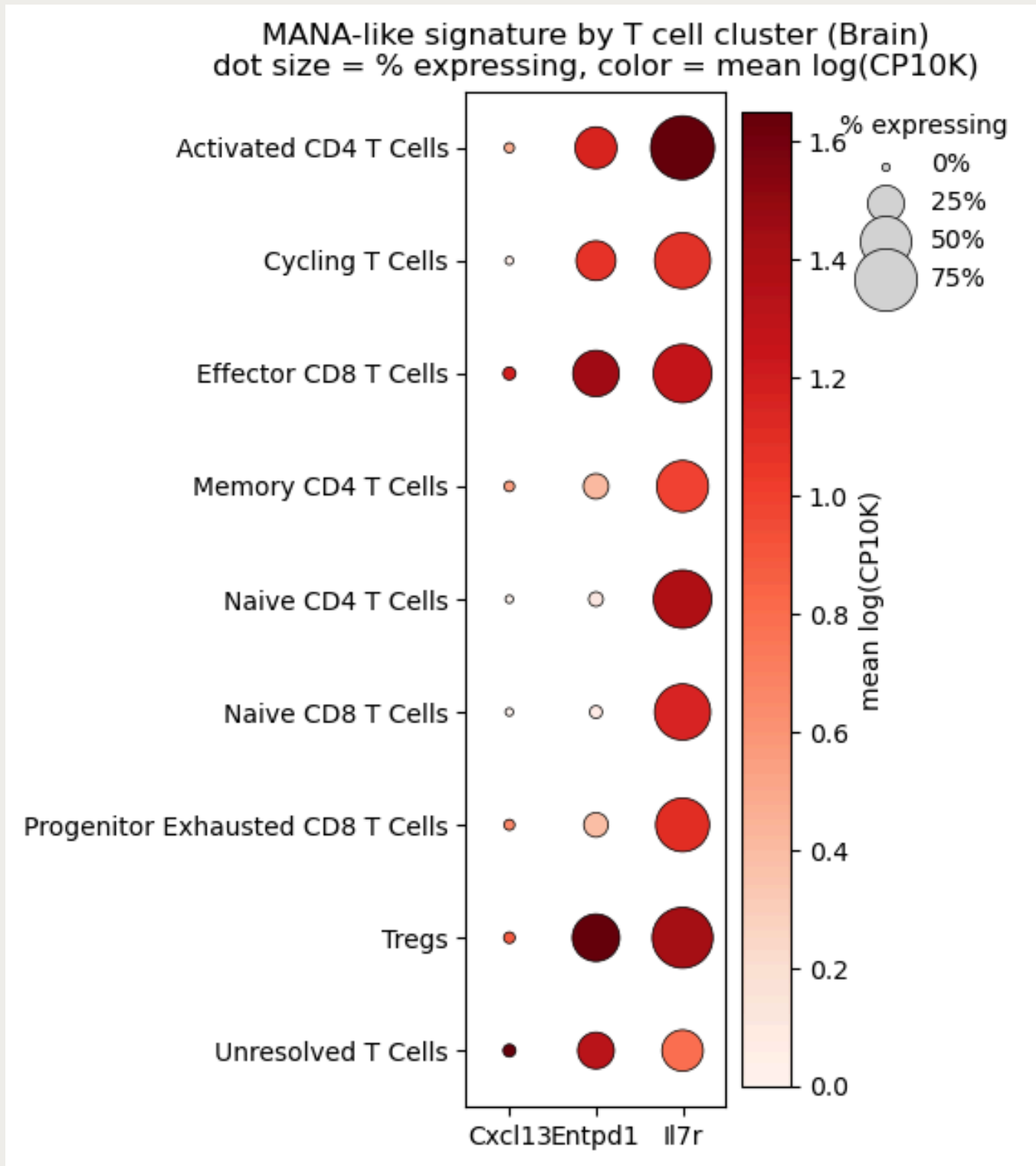
T cell subclustering splenocyte composition



MANA-specific CD8+ TIL.. express high levels of CXCL13 and CD39.. and low IL-7 receptor levels"
— Zeng et al., 2025

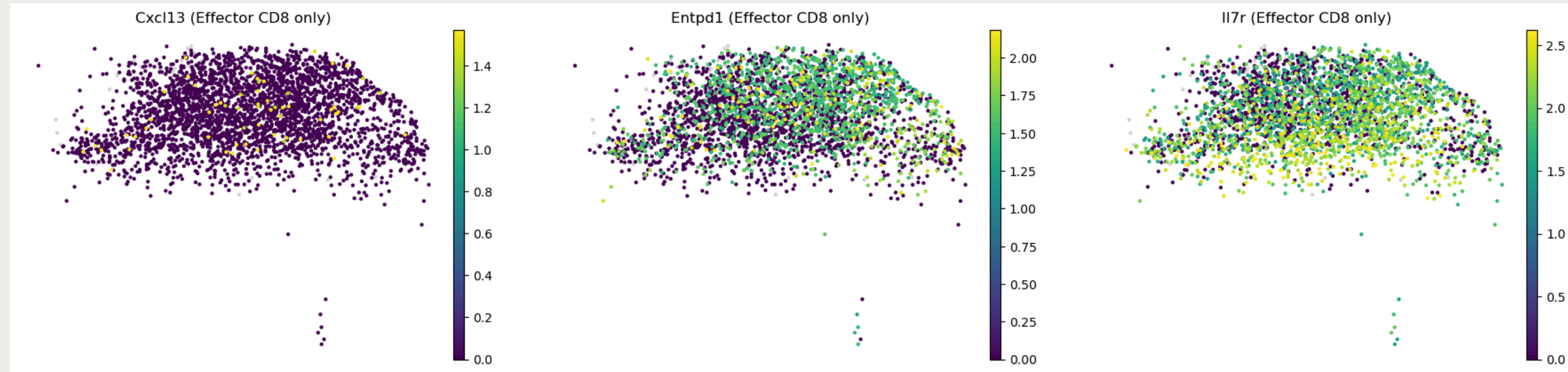
- Signature derived from human TIL across melanoma, lung cancer, and other cancer types, including one case with a virus-derived tumor antigen
- Question: does this pattern hold in our mouse brain tumor CD8/CD4 T cells?

MANAscore's signature doesn't reproduce (Cxcl13-hi, Entpd1-hi, Il7r-lo), no gene shows distinct spatial separation on the UMAP

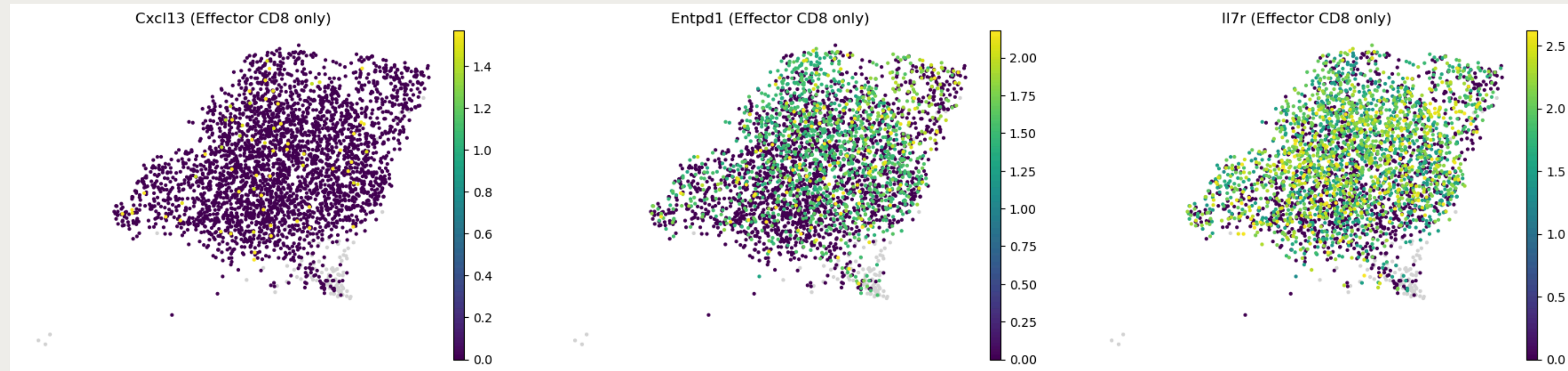


Zooming into effector T cells and subsetting CD8s alone doesn't recover the pattern

Effector T
cells zoomed
in (full T cell
UMAP)



CD8 T cells
only, zoomed
in on Effector
T cells



Thank You



Dr. Jeong

Johnathan Jia

Jeong Lab

